All India Aakash Test Series for NEET - 2025

TEST - 4 (Code-E)

Click here for Code-F Sol.

Test Date : 19/01/2025

ANSWERS

1. (1)	41. (1)	81. (1)	121. (3)	161. (1)
2. (3)	42. (2)	82. (3)	122. (4)	162. (2)
3. (3)	43. (3)	83. (1)	123. (3)	163. (2)
4. (3)	44. (4)	84. (3)	124. (2)	164. (3)
5. (3)	45. (1)	85. (1)	125. (2)	165. (1)
6. (2)	46. (2)	86. (2)	126. (3)	166. (4)
7. (3)	47. (3)	87. (3)	127. (1)	167. (2)
8. (4)	48. (1)	88. (4)	128. (4)	168. (1)
9. (2)	49. (2)	89. (3)	129. (3)	169. (2)
10. (2)	50. (1)	90. (2)	130. (4)	170. (2)
11. (2)	51. (2)	91. (1)	131. (2)	171. (1)
12. (2)	52. (3)	92. (3)	132. (4)	172. (4)
13. (1)	53. (2)	93. (1)	133. (3)	173. (3)
14. (2)	54. (3)	94. (4)	134. (2)	174. (3)
15. (1)	55. (2)	95. (4)	135. (4)	175. (2)
16. (2)	56. (4)	96. (4)	136. (2)	176. (4)
17. (1)	57. (4)	97. (2)	137. (2)	177. (2)
18. (2)	58. (2)	98. (2)	138. (3)	178. (3)
19. (3)	59. (4)	99. (2)	139. (1)	179. (1)
20. (3)	60. (4)	100. (1)	140. (1)	180. (2)
21. (2)	61. (3)	101. (4)	141. (4)	181. (1)
22. (2)	62. (3)	102. (4)	142. (4)	182. (3)
23. (2)	63. (2)	103. (3)	143. (2)	183. (1)
24. (2)	64. (1)	104. (4)	144. (2)	184. (3)
25. (4)	65. (1)	105. (2)	145. (2)	185. (1)
26. (2)	66. (3)	106. (2)	146. (3)	186. (4)
27. (4)	67. (2)	107. (2)	147. (1)	187. (2)
28. (3)	68. (3)	108. (1)	148. (2)	188. (2)
29. (3)	69. (1)	109. (3)	149. (2)	189. (1)
30. (1)	70. (3)	110. (3)	150. (4)	190. (3)
31. (3)	71. (2)	111. (3)	151. (3)	191. (2)
32. (3)	72. (1)	112. (2)	152. (2)	192. (1)
33. (1)	73. (2)	113. (3)	153. (4)	193. (3)
34. (3)	74. (2)	114. (2)	154. (1)	194. (3)
35. (1)	75. (4)	115. (2)	155. (3)	195. (4)
36. (2)	76. (3)	116. (1)	156. (4)	196. (3)
37. (2)	77. (2)	117. (3)	157. (2)	197. (2)
38. (1)	78. (4)	118. (2)	158. (2)	198. (4)
39. (3)	79. (1)	119. (1)	159. (1)	199. (1)
40. (1)	80. (1)	120. (2)	160. (3)	200. (3)

HINTS & SOLUTIONS

[PHYSICS]

SECTION-A

1. Answer (1)

Hint: Thermal resistance

$= \dfrac{\text{Length}}{\text{Thermal conductivity} \times \text{Area}}$

Sol.: $R_{\text{th}} = \dfrac{L}{kA}$

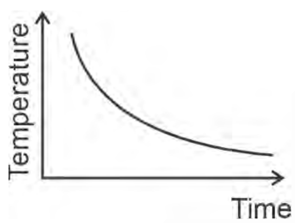
$AB = \dfrac{1}{L}$

$A^{-1}B^{-1} = L$

2. Answer (3)

Hint: Use Newton's law of cooling

Sol.:

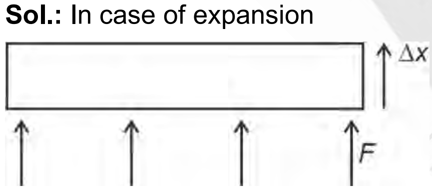


Rate of cooling is directly proportional to temperature difference between body and its surrounding. As temperature difference decreases, so time taken for cooling increases for the same drop in temperature.

3. Answer (3)

Hint: $W.D. = \vec{F} \cdot \vec{S}$, Force $F = PA$

Sol.: In case of expansion



$\therefore$ $W.D.$ is positive

4. Answer (3)

Hint: Use ideal gas equation; $PV = nRT$

Sol.: $\dfrac{P}{T} = \dfrac{nR}{V}$

$\tan 45^\circ = \dfrac{3(8.314)}{V}$

$V = 25\ \text{m}^3$

5. Answer (3)

Hint & Sol.: Carnot cycle is reversible engine and it consists of two isothermal and two adiabatic process.

6. Answer (2)

Hint: Adiabatic index, $\gamma = \dfrac{C_p}{C_v}$

Sol.: Slope of adiabatic curve = γ × slope of isothermal curve.

$\gamma = \dfrac{y}{x}$

7. Answer (3)

Hint: $\eta_{\text{max}} = 1 - \dfrac{T_L}{T_H}$

Sol.: Maximum efficiency = $1 - \dfrac{T_L}{T_H}$

$= 1 - \dfrac{300}{600} = 50\%$

8. Answer (4)

Hint: Total energy, $E = \dfrac{1}{2}kA^2$

Sol.: $E = \dfrac{1}{2}m\omega^2 a^2 \Rightarrow E \propto a^2$

$\dfrac{E'}{E} = \dfrac{\left(\dfrac{3}{4}a\right)^2}{a^2} \Rightarrow E' = \dfrac{9}{16}E$

9. Answer (2)

Hint: $\dfrac{C - 0}{100} = \dfrac{F - 32}{212 - 32}$

Sol.: $\dfrac{x}{100} = \dfrac{x - 32}{180} \Rightarrow x = -40$

10. Answer (2)

Hint: Use $\Delta x \times \dfrac{2\pi}{\lambda} = \Delta\phi$

Sol.: $\lambda = \dfrac{v}{f} = \dfrac{360}{600} = 0.6\ \text{m}$

Path difference $= \dfrac{\lambda}{2\pi}$ × phase difference

Path difference $= \dfrac{0.6}{2\pi} \times \dfrac{\pi}{3} = 0.10\ \text{m}$

11. Answer (2)

Hint & Sol.: Mean free path = $\dfrac{K_B T}{\sqrt{2}\pi P d^2} \Rightarrow \lambda \propto T$

Since temperature is halved, mean path will be halved.

12. Answer (2)

Hint: Energy of a molecule $= f\dfrac{1}{2}KT$, f is degree of freedom.

Sol.: Average energy of a molecule of an ideal monoatomic gas = $\dfrac{3}{2}KT$.

Average energy per molecule of a triatomic gas = 3 KT

13. Answer (1)

Hint: At constant pressure $Q = nC_P\Delta T$

Sol.: $\dfrac{\text{Work done}}{\text{Heat supplied}} = \dfrac{nR\Delta T}{nC_P\Delta T}$

$$\frac{W}{Q} = \frac{R}{C_P} = \frac{R}{\frac{5}{2}R} = \frac{2}{5}$$

14. Answer (2)

Hint & Sol.: In adiabatic expansion, internal energy of the system decreases.

Internal energy is a function of temperature.

15. Answer (1)

Hint: $\sin(A + B) = \sin A\cos B + \cos A\sin B$

Sol.: $x = A(\cos\pi t + \sin\pi t)$

$$= 4\sqrt{2}\left(\frac{1}{\sqrt{2}}\cos\pi t + \frac{1}{\sqrt{2}}\sin\pi t\right)$$

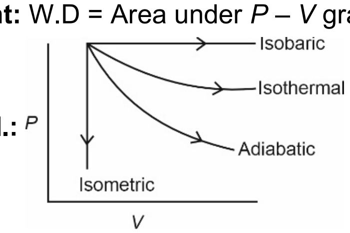
$$= 4\sqrt{2}\left(\sin\pi t\cos\frac{\pi}{4} + \cos\pi t\sin\frac{\pi}{4}\right)$$

$$= 4\sqrt{2}\sin\left(\pi t + \frac{\pi}{4}\right)$$

16. Answer (2)

Hint: W.D = Area under $P – V$ graph.

Sol.:



Work done is given by area under $P – V$ curve, hence work done in isobaric expansion is maximum.

17. Answer (1)

Hint: $W = P\Delta V = \Delta Q - \Delta U$

Sol.: $W = P\Delta V = nR\Delta T$ for isobaric process

$W = -\Delta U$ for adiabatic process

$P\Delta V = 0$ for isochoric process

$W = \Delta Q$ for Isothermal process

18. Answer (2)

Hint & Sol.: In stationary wave, nodes being permanently at rest, so no energy can be transmitted across them.

19. Answer (3)

Hint: $V = \dfrac{-(\text{coefficient of } t)}{(\text{coefficient of } x)}$

Sol.: $y = A\sin 2\pi\left(\dfrac{t}{T} - \dfrac{x}{\lambda}\right)$

Or

$y = A\sin(\omega t - kx)$ represents simple harmonic plane progressive wave travelling along $+x$ direction.

20. Answer (3)

Hint: $P\cdot E = \dfrac{1}{2}m\omega^2 y^2,\ K\cdot E = \dfrac{1}{2}m\omega^2(A^2 - y^2)$

$$\frac{1}{2}m\omega^2 y^2 = \frac{1}{2}m\omega^2(A^2 - y^2)$$

Sol.: $\dfrac{1}{2}m\omega^2 y^2 = \dfrac{1}{2}m\omega^2(A^2 - y^2)$

$$y = \frac{A}{\sqrt{2}} = 0.707A = 70\% \text{ of } A$$

21. Answer (2)

Hint: $v_{\text{rms}} = \sqrt{\dfrac{3RT}{M}}$

Sol.: $v_{\text{rms}} \propto \sqrt{T}$

$$\frac{v}{v'} = \frac{\sqrt{T}}{\sqrt{T'}}$$

$T' = T + 3T = 4T$

$v' = 2v$

22. Answer (2)

Hint: For polytropic process

$C = C_v + \dfrac{R}{1-N}$, PV^N = constant

Sol.: $C = C_V + \dfrac{R}{1-N}$

$R = \frac{3}{2}R + \frac{R}{1-N} \quad N = 3$

$\therefore$ PV^3 = constant

23. Answer (2)

Hint & Sol.: $v_{\text{rms}} = \sqrt{\frac{3RT}{M}}$, $P = \frac{1}{3}\rho(v_{\text{rms}})^2$,

$U = nC_vT$, Translational $K.E. = \frac{3}{2}K_BT$

24. Answer (2)

Hint: $(C_V)_{\text{mix}} = \frac{n_1C_{V_1} + n_2C_{V_2}}{n_1 + n_2}$

Sol.: $C_V = \frac{f}{2}R$

$C_{V_1} = \frac{5}{2}R, \quad n_1 = 2$

$C_{V_2} = \frac{3}{2}R, \quad n_2 = 4$

$(C_V)_{\text{mix}} = \frac{2 \times \frac{5}{2}R + 4 \times \frac{3}{2}R}{6} = \frac{11}{6}R$

25. Answer (4)

Hint: Average K.E $\propto$ Temperature

Sol.: $E = \frac{5}{2}kT$

For $\frac{E_{H_2}}{E_{O_2}} = \frac{\frac{5}{2}k\ 373}{\frac{5}{2}k\ 473}$

$E_{O_2} = 1.27E$

26. Answer (2)

Hint: Characteristics of standing waves

Sol.: Two travelling waves of same amplitude frequency and velocity moving in opposite directions are superposed, the phenomenon observed is standing waves.

27. Answer (4)

Hint: Efficiency of Carnot engine, $\eta = 1 - \frac{T_L}{T_H}$

Sol.: $0.4 = 1 - \frac{T_2}{T}$, $T_2 = 0.6T$

$0.5 = 1 - \frac{T_2}{T'}$, $0.5 = \frac{0.6T}{T'}$, $T' = \frac{6}{5}T$

28. Answer (3)

Hint: Thermal capacity $H = ms$

Sol.: $H = ms$

Since material is same

$\therefore \quad \frac{H_1}{H_2} = \frac{m_1}{m_2} = \frac{V_1}{V_2}$

$\frac{H_1}{H_2} = \left(\frac{R_1}{R_2}\right)^3 = \frac{1}{8}$

29. Answer (3)

Hint: By Wien's law $\lambda_m \propto \frac{1}{T}$

Sol.: $\frac{(\lambda_m)_1}{(\lambda_m)_2} = \frac{T_2}{T_1}$

$\frac{\lambda_m}{(\lambda_m)_2} = \frac{3000}{2000}$

$(\lambda_m)_2 = \frac{2}{3}\lambda_m$

30. Answer (1)

Hint: Ideal gas possess kinetic energy only.

Sol.: Average kinetic energy $\left(\frac{1}{2}m\bar{v}^2\right) = \frac{3}{2}k_BT$

$v_{\text{rms}} = (\bar{v}^2)^{1/2} = \sqrt{\frac{3k_BT}{m}}$

31. Answer (3)

Hint: Minimum time taken is from $\left(-\frac{A}{2} \text{ to } +\frac{A}{2}\right)$

Maximum time taken from $(0 - A)$

Sol.: Time taken will be minimum when it travels from $\left(-\frac{A}{2} \text{ to } \frac{A}{2}\right)$

$T_{\min} = \frac{T}{12} + \frac{T}{12} = \frac{T}{6} = 2$ s

$T_{\max} = \frac{T}{4} = 3$ s

$\frac{T_{\min}}{T_{\max}} = \frac{2}{3}$

32. Answer (3)

Hint: $v_{\max} = A\omega$

Sol.: $v_{\max} = A\omega$

$= 4(2) = 8$ m/s

33. Answer (1)

Hint: First resonance in a resonance tube is obtained for $l + e = \frac{\lambda}{4}$

Sol.: $l + e = \frac{\lambda}{4}$ ...(i)

$l' + e = \frac{3\lambda}{4}$ ...(ii)

Solving we get, l' = 31 cm

34. Answer (3)

Hint: Apply principle of calorimetry.

Sol.: Let mass of water and ice is equal to *m*.

Water (60°C) $\xrightarrow{Q_1}$ Water (0°C)

$Q_1 = m \times 1 \times 60 = 60m$ ...(i)

Ice (0°C) $\xrightarrow{Q_2}$ Water (0°C)

$Q_2 = m \times 80$ ...(ii)

Clearly, $Q_2 > Q_1$

Hence mixture will contain some ice at 0°C.

35. Answer (1)

Hint & Sol.: Displacement wave and pressure wave have phase difference equal to $\frac{\pi}{2}$ radians.

SECTION-B

36. Answer (2)

Hint: In string fixed at both ends,

3rd overtone = 4(fundamental frequency)

Sol.: Third overtone is 4th harmonic

$4f = 600$

$f = 150$ Hz

37. Answer (2)

Hint: In simple harmonic motion

Displacement $y = A\sin\omega t$

Acceleration $a = -\omega^2 y$

Velocity $v = \omega\sqrt{A^2 - y^2}$

Sol.: $v = \omega\sqrt{A^2 - y^2}$

$v^2 = A^2\omega^2 - \omega^2 y^2$

$\frac{v^2}{A^2\omega^2} + \frac{y^2}{A^2} = 1$

Equation of ellipse.

38. Answer (1)

Hint: Use equation $\left(\frac{dQ}{dt}\right)_A + \left(\frac{dQ}{dt}\right)_B + \left(\frac{dQ}{dt}\right)_C = 0$

Sol.: $KA\frac{(70 - T_0)}{l} + \frac{KA(70 - T_0)}{l} + \frac{KA(40 - T_0)}{l} = 0$

$T_0 = \frac{180}{3} = 60°C$

39. Answer (3)

Hint & Sol.: Standing wave is given by $y = 2A \sin kx \cos\omega t$

40. Answer (1)

Hint: Time period of spring pendulum is given by

$T = 2\pi\sqrt{\frac{m}{k}}$

Sol.: $T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{0.2}{80}} = 0.314$ s

41. Answer (1)

Hint: Use $v = \omega\sqrt{A^2 - x^2}$

Sol.: $v = \omega\sqrt{A^2 - x^2}$

$= 2\sqrt{(6)^2 - (2)^2}$

$= 2\sqrt{32} = 2 \times 5.65 = 11.3$ cm/s

42. Answer (2)

Hint: $\frac{dQ}{dt} = KA\frac{dT}{dx}$

Sol.: $\frac{\left(\frac{dQ}{dt}\right)_A}{\left(\frac{dQ}{dt}\right)_B} = \frac{K_1A_1\frac{dT}{l_1}}{K_2A_2\frac{dT}{l_2}}$

$\frac{4}{1} = \frac{K_1A_1}{K_2A_2}\frac{l_2}{l_1}$

$K_1A_1 = 8K_2A_2$

43. Answer (3)

Hint: Equation of travelling wave moving along negative *x*-axis is given by $y = A\sin(\omega t + kx)$

Sol.: $k = \frac{2\pi}{\lambda}$ $\therefore k = \frac{2\pi}{0.5} = 4\pi$

$y = 0.25\sin(\omega t + 4x)$

44. Answer (4)

Hint: Use ideal gas equation.

Sol.: $PV = nRT$

$n = \frac{PV}{RT}$

$n = \frac{1}{2} = 16$ g of O_2 = 1 g of H_2

45. Answer (1)

Hint: Time period of simple pendulum $T \propto \sqrt{\ell}$

Sol.: $T = 2\pi\sqrt{\frac{\ell}{g}} \Rightarrow T \propto \sqrt{\ell}$

$\frac{2}{T'} = \frac{1}{\sqrt{1-\frac{7}{16}}}$ $\frac{2}{T'} = \frac{4}{3}$

$T' = \frac{3}{2} = 1.5$ second

46. Answer (2)

Hint: $y = A\sin\omega t$

$v = \omega\sqrt{A^2 - y^2}$

Sol.: At $\frac{T}{12}$, $y = \frac{A}{2}$

$v = \omega\sqrt{A^2 - \frac{A^2}{4}} = \frac{\sqrt{3}A\omega}{2}$

47. Answer (3)

Hint & Sol.: Sound waves transfer both energy and momentum.

48. Answer (1)

Hint & Sol.: Volume of the cavity will increase while heating.

49. Answer (2)

Hint: Total degree of freedom of a diatomic gas is 5.

Sol.: $C_v = \frac{R}{\gamma - 1}$

$\gamma = \frac{7}{5}$

$C_v = \frac{R}{\frac{7}{5} - 1} = \frac{5}{2}R$

50. Answer (1)

Hint & Sol.: Work done in adiabatic process is given by $W = \frac{nR(T_1 - T_2)}{\gamma - 1}$

$= \frac{P_1V_1 - P_2V_2}{\gamma - 1}$

[CHEMISTRY]

SECTION-A

51. Answer (2)

Hint: $\Delta T_f = i\, K_f\, m$

Sol.: For P_1 solution;

im = $i \times \frac{20}{180} \times \frac{1000}{500} = 0.22$ m

$(\Delta T_f)_1 = 0.22\ K_f$

$(T_f)_1 = -0.22\ K_f$

For P_2 solution, im = 1 × $\frac{20}{60} \times \frac{1000}{500}$

= 0.66 m

$(\Delta T_f)_2 = 0.66\ K_f$

$(T_f)_2 = -0.66\ K_f$

For P_3 solution, im = 1 × $\frac{20}{342} \times \frac{1000}{500}$

= 0.12 m

$(\Delta T_f)_3 = 0.12\ K_f$

$(T_f)_3 = -0.12\ K_f$

Decreasing order of freezing point of given solutions is $P_3 > P_1 > P_2$.

52. Answer (3)

Hint: For negative deviation from Raoult's law, the interactions between A---A, B---B weaker than A---B.

Sol.: Due to intermolecular H-bonding, the solution of acetone and chloroform show negative deviation from Raoult's Law.

53. Answer (2)

Hint: $y_A = \frac{P_A^o X_A}{P_A^o X_A + P_B^o X_B}$

Sol.: $P_A = P_A^o X_A = 250$

$X_A = \frac{250}{500} = \frac{1}{2}$

$X_B = 1 - \frac{1}{2} = \frac{1}{2}$

$y_A = \frac{P_A^o X_A}{P_A^o X_A + P_B^o X_B}$

$= \frac{250}{250 + 200 \times \frac{1}{2}}$

$$y_A = \frac{250}{250+100} = \frac{250}{350} = \frac{25}{35} = \frac{5}{7}$$

$$y_B = 1 - \frac{5}{7} = \frac{2}{7}$$

Molar ratio = $\frac{y_A}{y_B} = \frac{5/7}{2/7} = \frac{5}{2}$

54. Answer (3)

Hint: Volume changes with change in temperature.

Sol.: Molality and % (w/w) terms are not related with volume therefore these are temperature independent.

55. Answer (2)

Hint: $P = K_H X$

Sol.: Higher the volume of K_H at a given temperature and pressure, the lower is the solubility of the gas in the liquid.

56. Answer (4)

Hint: Amount of dissolved oxygen is more in cold water than in warm water.

Sol.: Due to greater oxygen content in cold water, aquatic species are more comfortable in cold water than warm water. Solubility of gases in liquid increases with decrease in temperature.

57. Answer (4)

Hint: $\Delta T_b = i\, K_b\, m$

Sol.:

(1) 0.1 m KCl (aq)

$i = 2$

$\Delta T_b = 2 \times 0.1\, K_b = 0.2\, K_b$

(2) 0.20 m $C_6H_{12}O_6$ (aq)

$i = 1$

$\Delta T_b = 1 \times 0.2 \times K_b = 0.2\, K_b$

(3) 0.10 m $Al_2(SO_4)_3$ (aq)

$i = 5$

$\Delta T_b = 5 \times 0.10 \times K_b = 0.5\, K_b$

(4) 0.20 m K_2SO_4 (aq)

$i = 3$

$\Delta T_b = 3 \times 0.2 \times K_b = 0.6\, K_b$

58. Answer (2)

Hint: Azeotropes are binary mixture having the same composition in liquid and vapour phase and boil at a constant temperature.

Sol.: Nitric acid and water can form maximum boiling azeotrope.

59. Answer (4)

Hint: Value of K_f (molal depression constant) depends on the nature of solvent.

Sol.: $K_f = \frac{RT_f^2}{1000\,\Delta H}$

ΔH represents enthalpy of fusion.

So, K_f is independent of molality of the solution.

60. Answer (4)

Hint: $X_A = \frac{n_A}{n_A + n_B}$

Sol.: 2 m aqueous solution means that 2 mole solute present in 1 kg of water.

$$X = \frac{2}{2+\frac{1000}{18}} = \frac{2}{2+55.5} = \frac{2}{57.5} = 0.035$$

61. Answer (3)

Hint: $\pi = iCRT$

$i = 1$

Sol.: $2.57 \times 10^{-3} = \frac{\frac{2.52}{M}}{\frac{400}{1000}} \times 0.083 \times 300$

$$2.57 \times 10^{-3} = \frac{2.52 \times 1000}{M \times 400} \times 0.083 \times 300$$

$$M = \frac{2.52 \times 10 \times 0.083 \times 300}{2.57 \times 4 \times 10^{-3}}$$

$$= \frac{2.52 \times 0.083 \times 3 \times 10^{6}}{2.57 \times 4}$$

$= 61038\ g\ mol^{-1}$

62. Answer (3)

Sol.:

	$A(g)$	$\rightarrow$	$2B(g)$	+	$C(g)$
t = 0	0.1				
t = 115	0.1 – x		2x		x

$0.1 + 2x = 0.28$

$2x = 0.18$

$x = 0.09$

$$k = \frac{2.303}{115}\log\left(\frac{0.1}{0.1-0.09}\right)$$

$k = 2 \times 10^{-2}\ sec^{-1}$

63. Answer (2)

Hint: Slowest step is the rate determining step

Sol.:

(1) $A_3 \underset{}{\overset{\text{Fast}}{\rightleftharpoons}} A_2 + A$

$$K_{eq} = \frac{[A_2][A]}{[A_3]}$$

$$[A] = K_{eq} \times \frac{[A_3]}{[A_2]} \quad \ldots(1)$$

(2) $A_3 + A \xrightarrow{\text{slow}} 2A_2$

$$r = K[A_3][A]$$

$$= K[A_3]\,K_{eq}\frac{[A_3]}{[A_2]}$$

$$= K'[A_3]^2\,[A_2]^{-1}$$

Overall order = (2) + (–1) = 1

64. Answer (1)

Hint: $t_{1/2} \propto a^{1-n}$

Sol.: For zero order reaction

$t_{1/2} \propto a^{1-0} \propto a^1$

Means half-life of zero order reaction is directly proportional to initial concentration of the reactant.

65. Answer (1)

Hint: $r = \frac{-d\left[BrO_3^-\right]}{dt} = \frac{-1}{5}\frac{d[Br^-]}{dt} = -\frac{1}{6}\frac{d[H^+]}{dt}$

$= \frac{1}{3}\frac{d[Br_2]}{dt} = \frac{1}{3}\frac{d[H_2O]}{dt}$

Sol.: $\frac{\text{Rate of appearance of } H_2O}{\text{Rate of disappearance of } Br^-} = \frac{3r}{5r} = \frac{3}{5}$

66. Answer (3)

Hint: $t = \frac{2.303}{k}\log\frac{[A]_0}{[A]_t}$

Sol.: $t = \frac{2.303}{k}\log\frac{100}{100-99.99}$

$= \frac{2.303}{k}\log\frac{100}{0.01}$

$= \frac{2.303}{k}\log 10^4$

$= \frac{4 \times 2.303}{k}$

67. Answer (2)

Hint and Sol.: Order of reaction is applicable to elementary as well as complex reactions.

68. Answer (3)

Hint: Rate = $k[A]^x[B]^y$ where x and y are the order w.r.t. A and B

Sol.: $r = k[A]^x[B]^y$ …(i)

$2r = k[2A]^x[B]^y$ …(ii)

$8r = k[2A]^x[2B]^y$ …(iii)

From equation (i), (ii) and (iii) we get

x = 1, y = 2

Rate = $k[A]^1[B]^2$

69. Answer (1)

Hint: $k = Ae^{-E_a/RT}$

Sol.: $\ln k = \ln A - \frac{E_a}{RT}$

Slope = $\frac{-E_a}{R}$, Intercept = ln A

$$\frac{\text{Slope}}{\text{Intercept}} = \frac{-E_a/R}{\ln A} = -\frac{E_a}{R \times \ln A}$$

70. Answer (3)

Hint: Rate = $PZ_{AB}\,e^{-E_a/RT}$

Sol.: $e^{-E_a/RT}$ represents the fraction of molecules with energies equal to or greater than E_a.

71. Answer (2)

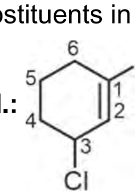
Hint: $CH{\equiv}C-CH_2-\underset{\displaystyle CH_3}{\underset{|}{C}}=CH_2$

Sol.: Number of sp^2 hybridised carbon atoms = 2

Number of π – bonds = 3

72. Answer (1)

Hint: Double bond have more priority than substituents in IUPAC naming.

Sol.: 

3-Chloro-1-methylcyclohex-1-ene.

73. Answer (2)

Hint: Carbocations are stabilized by +R, hyperconjugation and +I effect.

Sol.: Order of stability of carbocation 3° > 2° > 1°

74. Answer (2)

Hint and Sol: Thin layer chromatography is a type of adsorption chromatography where as paper chromatography is type of partition chromatography.

75. Answer (4)

Hint: Steam distillation is applied to separate substances which are steam volatile and are immiscible with water.

Sol.: Water and Aniline is separated by steam distillation method.

76. Answer (3)

Hint: Nitrogen, sulphur, halogens and phosphorus present in an organic compound are detected by “Lassaigne’s test”.

Sol.: Carbon and hydrogen are detected by heating the compounds with cupric oxide.

77. Answer (2)

Hint and Sol.:

	Column I (Compound)		**Column II (Common Name)**
a.	$C_6H_5OCH_3$	(i)	Anisole
b.	$C_6H_5COCH_3$	(ii)	Acetophenone
c.	$CH_3OCH_2CH_3$	(iii)	Ethyl methyl ether
d.	$C_6H_5NH_2$	(iv)	Aniline

78. Answer (4)

Hint: Polarization of σ bonded electron due to difference in electronegativity is known as inductive effect.

Sol.: Correct order of – I effect.

– CN > – F > – OH

79. Answer (1)

Hint: In resonance electrons are delocalised.

Sol.: $\overset{\ominus}{C}H_2 - \overset{O}{\overset{||}{C}} - CH_2 - CH_3 \leftrightarrow CH_2 = \overset{:\ddot{O}:^{\ominus}}{\overset{|}{C}} - CH_2 - CH_3$ are resonating structures.

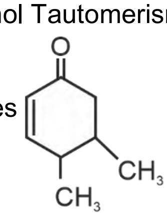
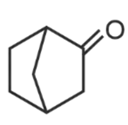
80. Answer (1)

Hint: Kjeldahl’s method is not applicable to compounds containing nitrogen in nitro and azo groups and nitrogen present in the ring (e.g Pyridine)

Sol.: Kjeldahl’s method is applicable for amine ($CH_3 - CH_2 - NH_2$).

81. Answer (1)

Hint: If a compound have enoliable hydrogen then it will show keto-enol Tautomerism.

Sol.: The molecules  and  have α – Hydrogen which can involve in Tautomerism.

82. Answer (3)

Hint: Structural isomers have same molecular formula but different structure.

Sol.: Possible structural isomers of $C_4H_{10}O$ are

$CH_3 - CH_2 - CH_2 - CH_2 - OH$

$CH_3 - \overset{H}{\overset{|}{\underset{CH_3}{\underset{|}{C}}}} - CH_2 - OH$

$CH_3 - CH_2 - \underset{OH}{\underset{|}{CH}} - CH_3$

$CH_3 - \overset{OH}{\overset{|}{\underset{CH_3}{\underset{|}{C}}}} - CH_3$

$CH_3 - \underset{CH_3}{\underset{|}{CH}} - O - CH_3$

$CH_3 - CH_2 - O - CH_2 - CH_3$

$CH_3 - CH_2 - CH_2 - O - CH_3$

83. Answer (1)

Hint: % of N = $\dfrac{28\ x}{22400 \times W} \times 100$

x is the volume of N_2 at STP.

Sol.: W = 0.30 g $\quad$ V_1 = 60 mL

P_1 = (715 – 15) = 700 mm of Hg

T_1 = 300 K

Now $\dfrac{P_1V_1}{T_1} = \dfrac{P_2V_2}{T_2}$

$$V_2 = \frac{P_1V_1}{T_1} \times \frac{T_2}{P_2} = \frac{700 \times 60 \times 273}{300 \times 760}$$

V_2 = 50.29 mL

$$\% \text{ of N} = \frac{28 \times 50.29}{22400 \times 0.30} \times 100$$

= 20.95%

84. Answer (3)

Hint: Nucleophiles are electron rich species.

Sol.: Nucleophiles attack at low electron density site and they are nucleus seeking species.

85. Answer (1)

Hint and Sol: Decomposition of gaseous ammonia on a hot platinum surface at high pressure is an example of zero order reaction

$$2NH_3(g) \xrightarrow[\text{Pt catalyst}]{1130\text{ K}} N_2(g) + 3H_2(g)$$

SECTION-B

86. Answer (2)

Hint and Sol.: Osmotic pressure is widely used to determine molar masses of proteins and polymers because pressure measured is around the room temperature and the molarity of solution is used instead of molality.

87. Answer (3)

Hint: For isotonic solution

$\boxed{\pi_1 = \pi_2}$

Sol.: $\pi_1 = \pi_2$

$C_1RT = C_2RT$

$C_1 = C_2$

$$\frac{6.84\times10}{342} = \frac{2\times10}{M}$$

$$2\times10^{-2} = \frac{2}{M}$$

$$M = \frac{1}{10^{-2}} = 100 \text{ g/mol}$$

88. Answer (4)

Hint: According to IUPAC nomenclature carboxylic acid and its derivatives are at higher priority.

Sol.: Correct order of priority of functional groups in IUPAC naming is

– COOH > – COCl > – OH > – NH_2

89. Answer (3)

Hint: $\Delta T_b = i K_b m$, $\Delta T_f = i K_f m$

Sol.: i = 1 for urea

$\Delta T_b = 101.024 - 100 = 1.024°C$

$1.024 = 0.512 \times m$

m = 2

$\Delta T_f = 1.86 \times 2 = 3.72°C$

$0 - T_f = 3.72°C$

$T_f = -3.72°C$

90. Answer (2)

Hint: $i = 1 - \alpha\left(1-\frac{1}{n}\right)$

Sol.: $1 - 0.3\left(1-\frac{1}{3}\right)$

$= 1 - 0.3 + 0.1 = 0.80$

91. Answer (1)

Hint: Relative lowering in vapour pressure = mole fraction of solute.

Sol.: $\frac{\Delta P}{P_A^o}\times100 = 40$

$x_B = 0.4$

$$x_B = \frac{n_B}{n_A + n_B} = \frac{w/60}{\frac{171}{114}+\frac{w}{60}} = 0.4$$

$$\frac{\frac{w}{60}}{\frac{171}{114}+\frac{w}{60}} = \frac{4}{10} = \frac{2}{5}$$

$$\frac{5w}{60} = 2\times\frac{171}{114}+\frac{2w}{60}$$

$$\frac{3w}{60} = 2\times\frac{171}{114}$$

$$w = 2\times20\times\frac{171}{114} = 60$$

92. Answer (3)

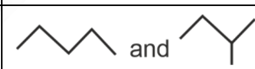
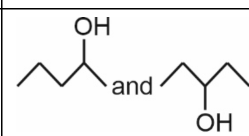
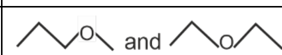
Hint: $\frac{r_{(t+\Delta t)}}{r_t} = (2)^{\frac{\Delta t}{10}}$

Sol.: $\frac{r_{25}}{r_{55}} = \frac{1}{2^{\frac{55-25}{10}}} = \frac{1}{2^3} = \frac{1}{8}$

93. Answer (1)

Hint: Structural isomers have same molecular formula but different structure.

Sol.:

	Column I (Structures)	**Column II (Type of isomers)**
a.		Chain isomers
b.		Position isomers
c.	$CH_3 - CH_2 - CHO$ and $CH_3 - CO - CH_3$	Functional group isomers
d.		Metamers

94. Answer (4)

Hint and Sol: Correct order of stability of free radical II > III > I

$\overset{}{C}H_2 = CH - \dot{C}H_2 \longleftrightarrow \dot{C}H_2 - CH = CH_2$

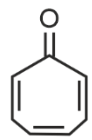
[two equivalent resonating structure]

Therefore, allylic radical is more stable than benzylic radical.

95. Answer (4)

Hint: Aromatic compounds containing benzene ring are known as benzenoids

Sol.:



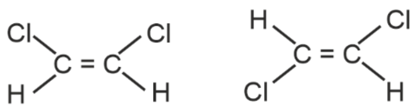
Tropone

Non-benzenoid aromatic compound

96. Answer (4)

Hint: If the two atoms or groups attached to each sp^2 hybridised carbon atom of alkene are different, they can show geometrical isomerism.

Sol.: Following molecules are two geometrical isomer



cis form trans form

97. Answer (2)

Hint: Percentage of halogen (X)

$$= \frac{\text{atomic mass of } X \times m_1 \times 100}{\text{molecular mass of } AgX \times m}$$

m_1 = mass of AgX

m = mass of organic compound

Sol.: m_1 = 1.128 g, m = 1 g

M(AgBr) = 188 g mol^{-1}

$$\% \text{ of Br} = \frac{80 \times 1.128 \times 100}{188 \times 1} = 48\%$$

98. Answer (2)

Hint and Sol: If nitrogen or sulphur is present in the compound, the sodium fusion extract is first boiled with concentrated nitric acid to decompose cyanide or sulphide of sodium formed during Lassaigne's test.

99. Answer (2)

Hint: $\log\frac{K_2}{K_1} = \frac{E_a}{2.303\,R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$

Sol.: $\log\frac{K_2}{K_1} = \frac{14000}{2.303\,R}\left[\frac{1}{400} - \frac{1}{500}\right]$

$\log\frac{K_2}{K_1} = 0.365$, so $\frac{K_2}{K_1} = 2.32$

$K_2 = 3.7 \times 10^8$

100. Answer (1)

Hint and Sol: Catalyst catalyse the spontaneous reactions but does not catalyses non-spontaneous reactions.

[BOTANY]

SECTION-A

101. Answer (4)

Hint: Increase in girth is caused by lateral meristems.

Sol.: Cork cambium is an example of secondary meristem responsible for secondary growth.

102. Answer (4)

Hint: Meristematic cells have a large conspicuous nucleus.

Sol.: Meristematic cells have a primary cell wall but large vacuole and secondary cell wall is absent.

103. Answer (3)

Hint: Floating respiration uses fat or carbohydrate as respiratory substrates.

Sol.: Protein is used as a substrate in protoplasmic respiration.

104. Answer (4)

Hint: S-shaped curve is observed in geometric growth.

Sol.: An elongating root at constant rate shows arithmetic growth.

Geometric growth is seen when a zygote is developed into embryo.

105. Answer (2)

Hint: Blackman gave the law of limiting factors in 1905.

Sol.: The first action spectrum of photosynthesis was described through the experiments performed by T.W. Engelmann.

106. Answer (2)

Hint: Microorganisms show a typical geometric growth curve.

Sol.: Sigmoid growth curve is seen in the microorganisms, when they have limited food and space.

107. Answer (2)

Hint: Redox equivalents in form of hydrogen atoms are removed and transferred to NAD^+.

Sol.: Two redox-equivalents are removed from one PGAL molecule during the conversion of PGAL to BPGA.

108. Answer (1)

Hint: Intrinsic factors work from inside the cells.

Sol.: Genetic factors and PGRs are the most important intrinsic factors controlling the developmental processes of the plant.

109. Answer (3)

Hint: Joseph Priestley concluded that the plants restore to the air whatever the breathing mouse and the burning candle remove.

Sol.: (i) Jan Ingenhousz showed oxygen was released in presence of sunlight by green parts of plants.

(ii) Julius von Sachs proved that glucose is produced during photosynthesis in the green parts of the plant.

(iii) Cornelius van Niel showed that oxygen evolved by green plants comes from water and not from carbon dioxide.

110. Answer (3)

Hint: There is one step in glycolysis where NADH + H^+ is formed from NAD^+; this is when 3-phosphoglyceraldehyde (PGAL) is converted to 1,3-bisphosphoglycerate.

Sol.: Total ATP produced from glycolysis:

2 NADH = 2 × 3 ATP = 6 ATP

4 ATP = 4 ATP

= 10 ATP (Glycolysis)

Total ATP produced from link reaction and Krebs' cycle

8 NADH = 3 × 8 = 24 ATP

2 $FADH_2$ = 2 × 2 = 4 ATP

2 ATP = 2 ATP

= 30 ATP

So, Total ATP = 40 ATP

111. Answer (3)

Hint: RuBP is the primary CO_2 acceptor in C_3 plants.

Sol.: The primary CO_2 acceptor in C_4 plants is phosphoenol pyruvate.

112. Answer (2)

Hint: During decarboxylation, CO_2 molecule is released.

Sol.: Decarboxylation does not occur during glycolysis and lactic acid fermentation.

113. Answer (3)

Hint: Citrate synthase is the enzyme of TCA cycle.

Sol.: (i) Aldolase in an enzyme of the glycolytic pathway.

(ii) ATP synthase is involved in oxidative phosphorylation.

(iii) Pyruvate dehydrogenase is an enzyme of link reaction.

114. Answer (2)

Hint: Bakanae disease (foolish seedling disease) in rice was caused by a fungal pathogen, *Gibberella fujikuroi*.

Sol.: Gibberellin was isolated from the fungus *Gibberella fujikuroi*.

115. Answer (2)

Hint: Splitting of water is associated with non-cyclic photophosphorylation.

Sol.: (i) Water is the electron donor in non-cyclic photophosphorylation.

(ii) Both PS II and PS I are the parts of non-cyclic photophosphorylation.

P_{680} is associated with PS II.

116. Answer (1)

Sol.: The light saturation occurs at 10% of the total sunlight available to the plants.

117. Answer (3)

Hint: Exponential growth shows a sigmoid growth curve.

Sol.: $W_1 = W_0e^{rt}$ is the correct equation for exponential growth.

118. Answer (2)

Hint: There is an equal number of utilization of ATP and NADPH in the reduction phase of C_3 cycle.

Sol.: There will be a requirement of 4 ATP and 4 NADPH in the reduction phase of Calvin cycle for two molecules of CO_2 fixed. The ratio obtained will be 1 : 1.

119. Answer (1)

Hint: When fats and proteins are the respiratory substrate, the value of RQ is less than one.

Sol.: The RQ of fatty acid, tripalmitin is 0.7.

120. Answer (2)

Hint: Chlorophyll *a* is the chief photosynthetic pigment.

Sol.: Since the action spectrum of photosynthesis corresponds closely to the absorption spectrum of chlorophyll *a*, it helped in concluding that chlorophyll *a* is the chief photosynthetic pigment.

121. Answer (3)

Hint: Increase in the number of female flowers in cucumber is done by ethylene.

Sol.: Ethylene helps in initiating germination of seeds, such as peanuts.

122. Answer (4)

Hint: Gibberellins stimulate the synthesis of hydrolytic enzymes, which helps in seed germination.

Sol.: ABA (Abscisic acid) inhibits gibberellin mediated amylase formation during germination of seed.

123. Answer (3)

Hint: In cyclic photophosphorylation only ATP is synthesized.

Sol.: In cyclic photophosphorylation P_{700} is the reaction centre.

124. Answer (2)

Hint: The first 6-carbon compound is formed by the condensation of acetyl CoA with oxaloacetic acid and H_2O.

Sol.: The first step of Krebs' cycle involves the synthesis of citric acid. This reaction is catalysed by citrate synthase.

125. Answer (2)

Hint: Redifferentiated cells lose their ability to divide.

Sol.: Secondary xylem and cork are the examples of redifferentiated tissues.

126. Answer (3)

Hint: Granum refers to the stack of thylakoids.

Sol.: Pigments that help in light reactions are embedded in the thylakoid membrane.

127. Answer (1)

Hint: Substrate-level phosphorylation refers to the direct formation of ATP or GTP by transferring a phosphate group from a high energy compound to an ADP or GDP molecule.

Sol.: In each turn of citric acid cycle, there is a single substrate level phosphorylation. For glucose molecule there will be occurrence of 2 substrate level phosphorylation reactions during the citric acid cycle.

128. Answer (4)

Hint: Removal of apical bud promotes the growth of lateral buds.

Sol.: Application of auxins, like IBA, to the cut portion of decapitated shoots, prevents the growth of side branches.

129. Answer (3)

Hint: The enzyme is involved in the synthesis of phosphoenol pyruvate.

Sol.: PEP synthetase is the cold sensitive enzyme of the C_4 pathway.

130. Answer (4)

Hint: The chief photosynthetic pigment in green plants is chlorophyll *a*.

Sol.: Reaction centre in P_{700} and P_{680} is formed by chlorophyll *a*.

131. Answer (2)

Hint: This hormone, along with auxins is responsible for callus formation.

Sol.: Cytokinins and auxins are involved in morphogenesis or organogenesis.

132. Answer (4)

Hint: Apical dominance is caused by auxins.

Sol.: Cytokinins help in the production of new leaves and chloroplast in leaves.

133. Answer (3)

Hint: This phytohormone is the only gaseous hormone.

Sol.: Cousins discovered that ripened oranges emitted a volatile substance, which caused early ripening of bananas kept nearby. Later on, this volatile substance was identified as ethylene.

134. Answer (2)

Hint: The enzyme is attached to the inner membrane of mitochondria.

Sol.: Succinate dehydrogenase is the enzyme which is a part of both Krebs' cycle and ETS.

135. Answer (4)

Hint: Appearance of different forms of leaves on the same plant is called heterophylly.

Sol.: Buttercup shows environmental heterophylly.

SECTION-B

136. Answer (2)

Hint: Substrate level phosphorylation is the direct formation of ATP, when phosphate is transferred from the substrate.

Sol.: There will be 2 substrate level phosphorylation reactions, one during glycolysis and one during Krebs' cycle when PEP is used as a substrate.

137. Answer (2)

Hint & Sol.: No other alternative substrates enter the pathway according to assumptions made for respiratory balance sheet.

138. Answer (3)

Hint: Gibberellins cause fruits like apple to elongate.

Sol.: Ethylene is a gaseous hormone, which is used to synchronize fruit-set in pineapple and thining of fruits like cotton and cherry.

139. Answer (1)

Hint: The electrons needed to replace those removed from PS I are provided by photosystem II.

Sol.: Splitting of water molecule is mainly responsible for the source of electrons, used in replacing the ones lost from the reaction centre of PS II.

140. Answer (1)

Hint: Chlorophyll *b* shows maximum absorption in blue region of light spectrum.

Sol.: Blue region of light spectrum lies in range of 400-500 nm wavelength. In this region Chlorophyll *b* shows maximum absorption.

141. Answer (4)

Hint: Respiratory quotient value for proteins is less than unity.

Sol.: Respiratory quotient (RQ) of organic acids like oxalic acid is more than unity.

142. Answer (4)

Hint: Double carboxylation is a characteristic feature of C_4 plants.

Sol.: Bell pepper is a C_3 plant.

143. Answer (2)

Hint: Abscission is shedding of leaves and fruits.

Sol.: Both ethylene and ABA promote shedding of leaves.

144. Answer (2)

Hint: Methionine is the precursor of this plant hormone.

Sol.: Ethylene is the PGR which elongates stem in deep water rice plant.

145. Answer (2)

Hint: Tryptophan acts as a precursor for auxins.

Sol.:

a. Acetyl CoA serves as a precursor for gibberellins.

b. Methionine is a precursor for ethylene.

c. Violaxanthin is precursor for forming abscisic acid.

146. Answer (3)

Hint: The wall of bundle sheath cells are impervious to gaseous exchange.

Sol.: Bundle sheath cells of C_4 plants are characterized by large number of chloroplast, thick walls impervious to gaseous exchange and no intercellular spaces.

147. Answer (1)

Hint: Charles Darwin and his son concluded that tip of coleoptile was the site of transmittable influence that caused the bending of the entire coleoptile.

Sol.: F.W. Went, isolated auxins from the tips of coleoptile of oat seedlings.

148. Answer (2)

Hint: Assimilation number is the amount of CO_2 (in gms) assimilated by 1 g of chlorophyll in an hour.

Sol.: Assimilation number shows a relationship between the chlorophyll and photosynthesis.

149. Answer (2)

Hint: ABA inhibits protein and RNA synthesis, so as to promote senescence of leaves.

Sol.: The first natural cytokinin was isolated from unripe maize grain and it was called zeatin.

150. Answer (4)

Hint: ABA is known as the stress hormone.

Sol.: Inhibitor B, abscission II and dormin were chemically similar inhibitors, collectively called ABA.

[ZOOLOGY]

SECTION-A

151. Answer (3)

Hint: Characteristic feature of spiny bodied animals.

Sol.: Water vascular system is present in echinoderms. Water canal system is present in poriferans. Water enters through minute pores (ostia) in the body wall into a central cavity, spongocoel, from where it goes out through the osculum.

Choanocytes or collar cells line the spongocoel and the canals.

152. Answer (2)

Hint: An aquatic annelid

Sol.: *Nereis*, *Pheretima* and *Hirudinaria* belong to the phylum Annelida. *Nereis*, an aquatic form, is dioecious, but earthworm (*Pheretima*) and leech (*Hirudinaria*) are monoecious.

Balanoglossus belongs to the phylum Hemichordata and is unsegmented.

153. Answer (4)

Hint: Aquatic chordate

Sol.: Heart is ventral and CNS is dorsal, hollow and single in chordates. *Balaenoptera* (Blue whale) is a mammal and belongs to the phylum Chordata.

Aplysia, *Fasciola* and *Anopheles* are non-chordates. Heart is dorsal and CNS is ventral, solid and double in non-chordates.

154. Answer (1)

Hint: Transparent and membranous wings

Sol.: The first pair of wings arises from mesothorax and the second pair of wings arises from metathorax. Forewings (mesothoracic) called tegmina are opaque dark and leathery and cover the hindwings when at rest. The hindwings are transparent, membranous and are used in flight.

155. Answer (3)

Hint: A thin and flexible membrane.

Sol.: The entire body of cockroach (*Periplaneta americana*) is covered by a hard chitinous exoskeleton (brown in colour). In each segment, exoskeleton has hardened plates called sclerites (tergites dorsally and sternites ventrally) that are joined to each other by a thin and flexible articular membrane (arthrodial membrane).

156. Answer (4)

Hint: Coelenterates and ctenophores

Sol.: An undifferentiated layer, mesoglea, is present in between the external ectoderm and internal endoderm in diploblastic animals, seen in coelenterates and ctenophores.

Coelenterates – *Physalia*, *Adamsia*, *Gorgonia*, *Meandrina*

Ctenophore – *Pleurobrachia*

Molluscs – *Aplysia, Dentalium, Chaetopleura*

Echinoderm – *Antedon*

Both molluscs and echinoderms are triploblastic animals.

157. Answer (2)

Hint: Feature of aschelminths

Sol.: Roundworms are bilaterally symmetrical, triploblastic and pseudocoelomate animals. Alimentary canal is complete with a well-developed muscular pharynx. An excretory tube removes body wastes from the body cavity through the excretory pore.

Malpighian tubules are excretory structures present in arthropods. Statocyst is present in arthropods.

Body is covered by a calcareous shell in molluscs.

158. Answer (2)

Hint: Larval form of echinoderms

Sol.: Coelenterates and ctenophores exhibit radial symmetry. Platyhelminths are the first group of animals that show bilateral symmetry. From platyhelminths onwards, organisms exhibit bilateral symmetry.

Adult echinoderms are radially symmetrical whereas larval form of echinoderms are bilaterally symmetrical.

Meandrina – Coelenterate

Asterias, Ophiura – Echinoderms

Ctenoplana – Ctenophore

159. Answer (1)

Hint: Body cavity is absent in acoelomates

Sol.: Given figure represents diagrammatic sectional view of coelomate animals.

The body cavity which is lined by mesoderm is called coelom. Animals possessing coelom are called coelomates, *e.g.*, annelids, molluscs, arthropods, echinoderms, hemichordates and chordates.

In pseudocoelomates, the body cavity is not lined by mesoderm. Instead, the mesoderm is present as scattered pouches in between the ectoderm and endoderm. *E.g.*, Aschelminthes.

160. Answer (3)

Hint: One of them causes uterine contraction

Sol.: Neurohypophysis (pars nervosa) also known as posterior pituitary, stores and releases two hormones called oxytocin and vasopressin, which are actually synthesized by the hypothalamus and are transported axonally to neurohypophysis.

PRL, TSH, ACTH, LH are released by anterior pituitary.

GnRH and somatostatin are released by hypothalamus.

161. Answer (1)

Hint: PTH is a hypercalcemic hormone.

Sol.: The parathyroid glands secrete a peptide hormone called parathyroid hormone (PTH). The secretion of PTH is regulated by the circulating levels of calcium ions. PTH acts on the bones and stimulates the process of bone resorption. Hypersecretion of PTH increases the bone resorption and can be a possible cause of weakening of bones.

162. Answer (2)

Hint: Peptide hormones

Sol.: Hormones which interact with membrane bound receptors normally do not enter the target cell, but generate second messengers (*e.g.*, cyclic AMP, IP_3, Ca^{++}, *etc.*) which in turn regulate cellular metabolism.

Peptide, polypeptide and protein hormones generate second messengers. *E.g.*, Insulin, glucagon, pituitary hormones, hypothalamic hormones, *etc.*, Also, endocrine cells present in different parts of the gastro-intestinal tract secrete four major peptide hormones, namely gastrin, secretin, CCK and GIP.

Estradiol and testosterone are steroid hormones.

163. Answer (2)

Hint: Associated with glycosuria

Sol.: Diabetes mellitus – Associated with loss of glucose through urine and formation of harmful compounds known as ketone bodies. It could be caused due to prolonged hyperglycemia.

Acromegaly – Caused due to excess secretion of growth hormone in adults especially in middle age.

Addison's disease – Caused due to underproduction of hormones secreted by the adrenal cortex.

Exopthalmic goitre – A form of hyperthyroidism, also known as Graves' disease.

164. Answer (3)

Hint: Cephalochordate

Sol.: In the members of subphylum Cephalochordata and class Chondrichthyes, notochord is persistent throughout life.

Cephalochordate – *Branchiostoma*

Cyclostomes – *Petromyzon*, *Myxine*

Chondrichthyes – *Scoliodon*, *Pristis*, *Carcharodon*

In urochordates (*e.g.*, *Ascidia*, *Doliolum*, *Salpa*) notochord is present only in larval tail.

165. Answer (1)

Hint: Absence of air bladder

Sol.:

	Chondrichthyes	**Osteichthyes**
(1)	Skin is tough, containing minute placoid scales.	Skin is covered with cycloid/ctenoid scales.
(2)	Gill slits are without operculum.	They have four pairs of gills which are covered by an operculum on each side.
(3)	Due to absence of air bladder, they have to swim constantly to avoid sinking.	Air bladder is present which regulates buoyancy.

166. Answer (4)

Hint: Organ involved in hearing.

Sol.: Frog has different types of sense organs, namely organs of touch (sensory papillae), taste (taste-buds), smell (nasal epithelium), vision (eyes) and hearing (tympanum with internal ears). Out of these, eyes and internal ears are well-organised structures and the rest are cellular aggregations around nerve endings.

167. Answer (2)

Hint: Present between forebrain and hindbrain

Sol.: In frog, the brain is divided into forebrain, midbrain and hindbrain. Forebrain includes olfactory lobes, paired cerebral hemispheres and an unpaired diencephalon.

The midbrain is characterized by a pair of optic lobes. Hindbrain consists of cerebellum and medulla oblongata.

The medulla oblongata passes out through the foramen magnum and continues into spinal cord, which is enclosed in the vertebral column.

168. Answer (1)

Hint: Capillaries are present in closed circulatory system.

Sol.: Cyclostomes have a sucking and circular mouth without jaws. They have an elongated body bearing 6-15 pairs of gill slits for respiration.

Circulation is of closed type. Cyclostomes are marine but migrate for spawning to freshwater. After spawning, within a few days, they die. Their larvae, after metamorphosis return to the ocean.

169. Answer (2)

Hint: *Obelia* shows alternation of generation

Sol.: Bioluminescence is a well-marked property in ctenophores.

Few cnidarians exhibit metagenesis (alternation of generation) *i.e.*, polyps produce medusae asexually and medusae form the polyps sexually (*e.g.*, *Obelia*).

In molluscs, body is covered by a calcareous shell and is unsegmented with a distinct head, muscular foot and visceral hump.

170. Answer (2)

Hint: Inhibitory hormone released from hypothalamus

Sol.: The hormones produced by hypothalamus are of two types, the releasing hormones (which stimulate secretion of pituitary hormones) and the inhibiting hormones (which inhibit secretions of pituitary hormones). GnRH stimulates the pituitary synthesis of gonadotrophins.

On the other hand, somatostatin from the hypothalamus inhibits the release of growth hormone from the pituitary. PRL and FSH are released by anterior pituitary.

171. Answer (1)

Hint: Thyroxine is an iodothyronine.

Sol.: Hormones which interact with intracellular receptors (mostly nuclear receptors), regulate gene expression or chromosome function by the interaction of hormone-receptor complex with the genome.

E.g., steroid hormones, iodothyronines, *etc*.

Steroid hormones – (Cortisol, testosterone, estradiol and progesterone).

Iodothyronine – (Thyroxine).

172. Answer (4)

Hint: Present at the junction of midgut and hindgut

Sol.: In cockroach, gizzard or proventriculus helps in grinding of the food particles. A ring of 6-8 blind tubules called hepatic or gastric caeca is present at the junction of foregut and midgut, which secrete digestive juice.

At the junction of midgut and hindgut, is present another ring of 100-150 yellow coloured thin filaments called Malpighian tubules.

173. Answer (3)

Hint: Discharged during copulation

Sol.: In male cockroach, the sperms are stored in the seminal vesicles and are glued together in the form of bundles called spermatophores. On an average, female cockroach produces 9-10 oothecae, each containing 14-16 eggs. The development of *P. americana* is paurometabolous, meaning there is development through nymphal stage. The nymph looks very much like adults. The next to last nymphal stage has wing pads but only adult cockroaches have wings.

In cockroach, three ganglia lie in the thorax and six in the abdomen.

174. Answer (3)

Hint: Exclude the specialised venous connections

Sol.: In frogs, partially digested food called chyme is passed from stomach to the first part of the small intestine, the duodenum. The duodenum receives bile from gall bladder and pancreatic juices from the pancreas through a common bile duct.

Special venous connection between liver and intestine as well as the kidney and lower parts of the body are present in frogs. The former is termed as hepatic portal system and latter is called renal portal system.

175. Answer (2)

Hint: Vasa efferentia opens into Bidder's canal

Sol.: In frog, male reproductive organs consist of a pair of yellowish ovoid testes, which are found adhered to the upper part of the kidneys by mesorchium.

Vasa efferentia are 10-12 in number that arise from testes. They enter the kidneys on their side and open into Bidder's canal. Finally, it communicates with the urinogenital duct that comes out of the kidneys and open into the cloaca. Here, the ureters act as urinogenital duct. Urinary bladder is present ventral to the rectum which also opens in the cloaca.

The cloaca is a small, median chamber that is used to pass faecal matter, urine and sperms to the exterior.

Fat bodies do not open into cloaca.

176. Answer (4)

Hint: Possesses three – chambered heart

Sol.: Arthropods, hemichordates and molluscs (except cephalopods) possess an open circulatory system. Annelids (except leech) and chordates have closed circulatory system.

Periplaneta americana and *Locusta* – Arthropods

Saccoglossus – Hemichordate

Rana – Chordate

177. Answer (2)

Hint: *Hemidactylus* is a reptile.

Sol.:

Scientific name	Common name
Antedon	Sea lily
Dentalium	Tusk shell
Ornithorhynchus	Platypus
Hemidactylus	Wall lizard

178. Answer (3)

Hint: Acts on the thyroid follicles

Sol.: In humans, each thyroid follicle is composed of follicular cells, enclosing a cavity. These follicular cells synthesize two hormones, tetraiodothyronine or thyroxine (T4) and triiodothyronine (T3).

Iodine is essential for the normal rate of hormone synthesis in the thyroid gland. Deficiency of iodine in our diet results in hypothyroidism and enlargement of the thyroid gland is called goitre. During goitre, TSH levels increase which stimulates thyroid gland to synthesize more thyroxine.

TSH stimulates the synthesis and secretion of thyroid hormones from the thyroid gland.

179. Answer (1)

Hint: Cause for osteoporosis in females

Sol.: Decrease in the oestrogen levels can lead to increased chances of fracture in old age in females.

Low pitch is seen in males as compared to females. Testosterone affects the male libido and maturation of epididymis. Estrogen is secreted from growing follicles and corpus luteum in females. Glucocorticoids, particularly cortisol, produces anti-inflammatory reactions and suppresses the immune response.

180. Answer (2)

Hint: Also known as sea walnuts

Sol.: Ctenophores are exclusively marine. Many organisms in the phylum Platyhelminthes, Annelida and Mollusca are freshwater.

181. Answer (1)

Hint: Oxytocin

Sol.: The adrenal cortex secretes many hormones, commonly called corticoids. The corticoids, which are involved in carbohydrate metabolism are called glucocorticoids. Glucocorticoids stimulate gluconeogenesis, lipolysis and proteolysis and inhibit cellular uptake and utilisation of amino acids.

The α-cells of islets of Langerhans secrete a hormone called glucagon, while the β-cells secrete insulin. Glucagon is a peptide hormone, and plays an important role in maintaining the normal blood glucose levels. Glucagon acts mainly on the liver cells (hepatocytes) and stimulates glycogenolysis resulting in an increased blood sugar.

182. Answer (3)

Hint: Mantle cavity

Sol.: In phylum Mollusca, the body of animals is covered by a calcareous shell and is unsegmented with a distinct head, muscular foot and visceral hump. A soft and spongy layer of skin forms a mantle over the visceral hump. The space between the hump and the mantle is called mantle cavity in which feather-like gills are present.

The anterior head region has sensory tentacles. The mouth contains a file-like rasping organ for feeding called radula.

183. Answer (1)

Hint: Ventral side is the lower side in *Asterias*

Sol.: In echinoderms (*e.g.*, *Asterias*), digestive system is complete with mouth on the lower (ventral) side and anus on the upper (dorsal) side.

184. Answer (3)

Hint: Aschelminths

Sol.: Female roundworms are longer than males. Tapeworm, liver fluke and earthworm are monoecious and do not show sexual dimorphism.

185. Answer (1)

Hint: Organism in which tracheoles are present

Sol.: Proboscis gland is an excretory structure in *Balanoglossus*.

Protonephridia/flame cells are the excretory structure in platyhelminths. *Anopheles* belongs to the phylum Arthropoda and has Malpighian tubules as the excretory structures.

SECTION-B

186. Answer (4)

Hint: Function of MSH

Sol.: The pineal gland is located on the dorsal side of the forebrain.

Pineal gland secretes a hormone called melatonin.

Melatonin plays a very important role in the regulation of 24-hour rhythm of our body. It also helps in maintenance of body temperature.

In addition, melatonin also influences metabolism, pigmentation, the menstrual cycle as well as our defense capability. A hormone called MSH secreted from pars intermedia also regulates pigmentation.

187. Answer (2)

Hint: Has flame cells for excretion

Sol.: In poriferans, (*e.g.*, *Sycon*), cells are arranged as loose cell aggregates.

Tissue level of body organisation is seen in coelenterates (*e.g.*, *Pennatula*).

Incomplete digestive system means presence of a single opening which acts as both mouth and anus. Platyhelminths (*e.g.*, *Planaria*) has incomplete digestive system.

Enterobius vermicularis is an oviparous aschelminth.

188. Answer (2)

Hint: Acts on the melanocytes

Sol.: Pars intermedia secretes only one hormone called melanocyte stimulating hormone (MSH).

ACTH stimulates the synthesis and secretion of steroid hormones called glucocorticoids from the adrenal cortex. Both LH and FSH stimulate gonadal activity and are called gonadotropins.

In males, LH stimulates the synthesis and secretion of hormones called androgens from testis. In females, LH induces ovulation of fully mature follicle (Graafian follicles) and maintains the corpus luteum.

189. Answer (1)

Hint: Secreted by corpus luteum

Sol.: Progesterone, a steroid hormone supports pregnancy. Progesterone also acts on the mammary glands and stimulates the formation of alveoli (sac-like structures which store milk) and milk secretion.

Estrogen stimulates growth and development of female accessory sex organs. Prolactin regulates the growth of the mammary glands and formation of milk in them. Oxytocin acts on the smooth muscles of human female body and is a milk ejecting hormone.

190. Answer (3)

Hint: Additional chambers in digestive tract

Sol.: The characteristic feature of birds are the presence of feathers. The forelimbs are modified into wings. The hindlimbs generally has scales and are modified for walking, swimming or clasping the tree branches.

Skin is dry without glands except the oil gland at the base of the tail. Endoskeleton is fully ossified (bony) and the long bones are hollow with air cavities (pneumatic).

The digestive tract of birds and cockroaches have additional chambers, the crop and gizzard.

191. Answer (2)

Hint: Cyclostomes are jawless vetebrates.

Sol.: Agnatha and Gnathostomata are divisions of subphylum Vertebrata. Cyclostomata is a class under division Agnatha. Pisces and Tetrapoda are superclasses of division Gnathostomata.

192. Answer (1)

Hint: Ventricle of frog pumps out mixed blood.

Sol.: In frog, the heart is three-chambered, two atria and one ventricle and is covered by a membrane called pericardium. A triangular structure called sinus venosus joins the right atrium. It receives blood through major veins called vena cava. The ventricle opens into a sac-like conus arteriosus on the ventral side of the heart.

193. Answer (3)

Hint: Neck and tail are absent in frogs.

Sol.: Body of frog is divided into head and trunk. Tail is absent. The hindlimbs end in five webbed digits and they are larger and muscular than forelimbs that end in four digits.

Skin of frog is always in a moist condition. Scales are absent.

The colour of dorsal side of the body is generally olive green with dark irregular spots. On the ventral side, the skin is uniformly pale yellow.

Heart of frog is myogenic.

194. Answer (3)

Hint: ADH acts on DCT

Sol.: Vasopressin, released by neurohypophysis acts on kidneys and stimulates reabsorption of water and electrolytes by the distal renal tubules and thereby reduces the loss of water through urine (diuresis). Hence, it is also called anti-diuretic hormone (ADH).

195. Answer (4)

Hint: Thymus

Sol.: The thymus plays a major role in the development of the immune system. This gland secretes the peptide hormones called thymosins. Thymosins play a major role in the differentiation of T-lymphocytes, which provide cell-mediated immunity. In addition, thymosins also promote production of antibodies, to provide humoral immunity. Thymus is degenerated in old individuals.

196. Answer (3)

Hint: Catecholamines

Sol.: Antagonistic hormones produce opposite effects.

The adrenal medulla secretes two hormones called adrenaline or epinephrine and noradrenaline or norepinephrine. These are rapidly secreted in response to stress of any kind and during emergency situations.

Glucagon increases blood glucose levels whereas insulin decreases blood glucose levels.

Aldosterone leads to an increase in blood pressure whereas ANF can cause vasodilation and thereby, decrease the blood pressure.

PTH is a hypercalcemic hormone. Along with TCT, it plays a significant role in calcium balance in the body.

197. Answer (2)

Hint: Endorphins are pain relievers produced by the brain

Sol.: Endocrine cells present in different parts of the gastro-intestinal tract secrete four major peptide hormones; namely gastrin, secretin, CCK and GIP.

Gastrin acts on the gastric glands and stimulates the secretion of HCl and pepsinogen.

Secretin acts on the exocrine pancreas and stimulates the secretion of water and bicarbonate ions.

CCK acts on both pancreas and gall bladder and stimulates the secretion of pancreatic enzymes and bile juice, respectively. GIP inhibits gastric secretion and motility.

Endorphins are neuropeptides that block the perception of pain.

198. Answer (4)

Hint: Opens into the genital chamber

Sol.: Cockroaches are dioecious and both sexes have well-developed reproductive organs.

Male reproductive system consists of a pair of testes, one lying on each lateral side in the 4^{th} to 6^{th} abdominal segments. A characteristic mushroom-shaped gland is present in the 6^{th} to 7^{th} abdominal segments which functions as an accessory reproductive gland.

The female reproductive system consists of two large ovaries, lying laterally in the 2^{nd} to 6^{th} abdominal segments. A pair of spermatheca is present in the 6^{th} segment which opens into the genital chamber.

199. Answer (1)

Hint: Cockroach has nocturnal vision.

Sol.: In cockroach, each eye consists of about 2000 hexagonal ommatidia. With the help of several ommatidia, a cockroach can receive several images of an object. This kind of vision is known as mosaic vision with more sensitivity but less resolution, being common during night (hence called nocturnal vision).

200. Answer (3)

Hint: Labium is unpaired

Sol.: In cockroach, the mouthparts consist of a labrum (upper lip), a pair of mandibles, a pair of maxillae and a labium (lower lip).

Respiration takes place by network of tracheae. Tracheae opens outside with spiracles.

There are ten segments in abdomen.

Determinate and radial cleavage takes place during embryonic development.

❑ ❑ ❑

All India Aakash Test Series for NEET - 2025

TEST - 4 (Code-F)

Click here for Code-E Sol.

Test Date : 19/01/2025

ANSWERS

1. (1)	41. (1)	81. (2)	121. (2)	161. (2)
2. (3)	42. (4)	82. (3)	122. (2)	162. (3)
3. (3)	43. (3)	83. (2)	123. (3)	163. (3)
4. (3)	44. (2)	84. (3)	124. (2)	164. (4)
5. (3)	45. (1)	85. (2)	125. (3)	165. (1)
6. (2)	46. (1)	86. (1)	126. (3)	166. (2)
7. (3)	47. (3)	87. (2)	127. (3)	167. (2)
8. (4)	48. (1)	88. (2)	128. (1)	168. (1)
9. (2)	49. (2)	89. (2)	129. (2)	169. (2)
10. (2)	50. (2)	90. (4)	130. (2)	170. (4)
11. (2)	51. (1)	91. (4)	131. (2)	171. (1)
12. (2)	52. (3)	92. (4)	132. (4)	172. (3)
13. (1)	53. (1)	93. (1)	133. (3)	173. (2)
14. (2)	54. (3)	94. (3)	134. (4)	174. (2)
15. (1)	55. (1)	95. (1)	135. (4)	175. (1)
16. (2)	56. (1)	96. (2)	136. (4)	176. (3)
17. (1)	57. (1)	97. (3)	137. (2)	177. (1)
18. (2)	58. (4)	98. (4)	138. (2)	178. (2)
19. (3)	59. (2)	99. (3)	139. (1)	179. (2)
20. (3)	60. (3)	100. (2)	140. (3)	180. (4)
21. (1)	61. (4)	101. (4)	141. (2)	181. (3)
22. (3)	62. (2)	102. (2)	142. (2)	182. (1)
23. (1)	63. (2)	103. (3)	143. (2)	183. (4)
24. (3)	64. (1)	104. (4)	144. (4)	184. (2)
25. (3)	65. (2)	105. (2)	145. (4)	185. (3)
26. (1)	66. (3)	106. (4)	146. (1)	186. (3)
27. (3)	67. (1)	107. (3)	147. (1)	187. (1)
28. (3)	68. (3)	108. (4)	148. (3)	188. (4)
29. (4)	69. (2)	109. (1)	149. (2)	189. (2)
30. (2)	70. (3)	110. (3)	150. (2)	190. (3)
31. (4)	71. (1)	111. (2)	151. (1)	191. (4)
32. (2)	72. (1)	112. (2)	152. (3)	192. (3)
33. (2)	73. (2)	113. (3)	153. (1)	193. (3)
34. (2)	74. (3)	114. (4)	154. (3)	194. (1)
35. (2)	75. (3)	115. (3)	155. (1)	195. (2)
36. (1)	76. (4)	116. (2)	156. (2)	196. (3)
37. (2)	77. (4)	117. (1)	157. (1)	197. (1)
38. (1)	78. (2)	118. (2)	158. (3)	198. (2)
39. (3)	79. (4)	119. (3)	159. (2)	199. (2)
40. (2)	80. (4)	120. (1)	160. (4)	200. (4)

HINTS & SOLUTIONS

[PHYSICS]

SECTION-A

1. Answer (1)

Hint: Thermal resistance

$$= \frac{\text{Length}}{\text{Thermal conductivity} \times \text{Area}}$$

Sol.: $R_{\text{th}} = \frac{L}{kA}$

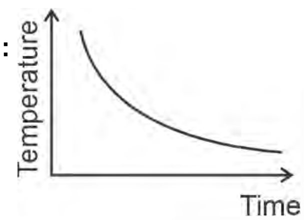
$AB = \frac{1}{L}$

$A^{-1}B^{-1} = L$

2. Answer (3)

Hint: Use Newton's law of cooling

Sol.:

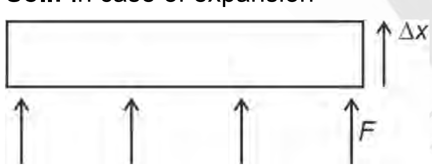


Rate of cooling is directly proportional to temperature difference between body and its surrounding. As temperature difference decreases, so time taken for cooling increases for the same drop in temperature.

3. Answer (3)

Hint: $W.D. = \vec{F} \cdot \vec{S}$, Force $F = PA$

Sol.: In case of expansion



$\therefore$ *W.D.* is positive

4. Answer (3)

Hint: Use ideal gas equation; $PV = nRT$

Sol.: $\frac{P}{T} = \frac{nR}{V}$

$\tan 45^\circ = \frac{3(8.314)}{V}$

$V = 25\ \text{m}^3$

5. Answer (3)

Hint & Sol.: Carnot cycle is reversible engine and it consists of two isothermal and two adiabatic process.

6. Answer (2)

Hint: Adiabatic index, $\gamma = \frac{C_p}{C_v}$

Sol.: Slope of adiabatic curve = γ × slope of isothermal curve.

$\gamma = \frac{y}{x}$

7. Answer (3)

Hint: $\eta_{\text{max}} = 1 - \frac{T_L}{T_H}$

Sol.: Maximum efficiency = $1 - \frac{T_L}{T_H}$

$= 1 - \frac{300}{600} = 50\%$

8. Answer (4)

Hint: Total energy, $E = \frac{1}{2}kA^2$

Sol.: $E = \frac{1}{2}m\omega^2 a^2 \Rightarrow E \propto a^2$

$$\frac{E'}{E} = \frac{\left(\frac{3}{4}a\right)^2}{a^2} \quad \Rightarrow \quad E' = \frac{9}{16}E$$

9. Answer (2)

Hint: $\frac{C-0}{100} = \frac{F-32}{212-32}$

Sol.: $\frac{x}{100} = \frac{x-32}{180} \Rightarrow x = -40$

10. Answer (2)

Hint: Use $\Delta x \times \frac{2\pi}{\lambda} = \Delta\phi$

Sol.: $\lambda = \frac{v}{f} = \frac{360}{600} = 0.6\ \text{m}$

Path difference $= \frac{\lambda}{2\pi}$ × phase difference

Path difference $= \frac{0.6}{2\pi} \times \frac{\pi}{3} = 0.10\ \text{m}$

11. Answer (2)

    Hint & Sol.: Mean free path = $\dfrac{K_B T}{\sqrt{2}\pi P d^2} \Rightarrow \lambda \propto T$

    Since temperature is halved, mean path will be halved.

12. Answer (2)

    Hint: Energy of a molecule $= f\dfrac{1}{2}KT$, f is degree of freedom.

    Sol.: Average energy of a molecule of an ideal monoatomic gas = $\dfrac{3}{2}KT$.

    Average energy per molecule of a triatomic gas = 3 KT

13. Answer (1)

    Hint: At constant pressure $Q = nC_P\Delta T$

    Sol.: $\dfrac{\text{Work done}}{\text{Heat supplied}} = \dfrac{nR\Delta T}{nC_P\Delta T}$

    $\dfrac{W}{Q} = \dfrac{R}{C_P} = \dfrac{R}{\frac{5}{2}R} = \dfrac{2}{5}$

14. Answer (2)

    Hint & Sol.: In adiabatic expansion, internal energy of the system decreases.

    Internal energy is a function of temperature.

15. Answer (1)

    Hint: $\sin(A + B) = \sin A\cos B + \cos A\sin B$

    Sol.: $x = A(\cos\pi t + \sin\pi t)$

    $= 4\sqrt{2}\left(\dfrac{1}{\sqrt{2}}\cos\pi t + \dfrac{1}{\sqrt{2}}\sin\pi t\right)$

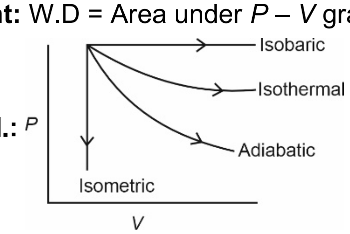
    $= 4\sqrt{2}\left(\sin\pi t\cos\dfrac{\pi}{4} + \cos\pi t\sin\dfrac{\pi}{4}\right)$

    $= 4\sqrt{2}\sin\left(\pi t + \dfrac{\pi}{4}\right)$

16. Answer (2)

    Hint: W.D = Area under $P - V$ graph.

    Sol.:



    Work done is given by area under $P - V$ curve, hence work done in isobaric expansion is maximum.

17. Answer (1)

    Hint: $W = P\Delta V = \Delta Q - \Delta U$

    Sol.: $W = P\Delta V = nR\Delta T$ for isobaric process

    $W = -\Delta U$ for adiabatic process

    $P\Delta V = 0$ for isochoric process

    $W = \Delta Q$ for Isothermal process

18. Answer (2)

    Hint & Sol.: In stationary wave, nodes being permanently at rest, so no energy can be transmitted across them.

19. Answer (3)

    Hint: $V = \dfrac{-(\text{coefficient of } t)}{(\text{coefficient of } x)}$

    Sol.: $y = A\sin 2\pi\left(\dfrac{t}{T} - \dfrac{x}{\lambda}\right)$

    Or

    $y = A\sin(\omega t - kx)$ represents simple harmonic plane progressive wave travelling along $+x$ direction.

20. Answer (3)

    Hint: $P\cdot E = \dfrac{1}{2}m\omega^2 y^2$, $K\cdot E = \dfrac{1}{2}m\omega^2(A^2 - y^2)$

    $\dfrac{1}{2}m\omega^2 y^2 = \dfrac{1}{2}m\omega^2(A^2 - y^2)$

    Sol.: $\dfrac{1}{2}m\omega^2 y^2 = \dfrac{1}{2}m\omega^2(A^2 - y^2)$

    $y = \dfrac{A}{\sqrt{2}} = 0.707A = 70\%$ of A

21. Answer (1)

    Hint & Sol.: Displacement wave and pressure wave have phase difference equal to $\dfrac{\pi}{2}$ radians.

22. Answer (3)

    Hint: Apply principle of calorimetry.

    Sol.: Let mass of water and ice is equal to m.

    Water (60°C) $\xrightarrow{Q_1}$ Water (0°C)

    $Q_1 = m \times 1 \times 60 = 60m$ …(i)

    Ice (0°C) $\xrightarrow{Q_2}$ Water (0°C)

    $Q_2 = m \times 80$ …(ii)

    Clearly, $Q_2 > Q_1$

    Hence mixture will contain some ice at 0°C.

23. Answer (1)

    Hint: First resonance in a resonance tube is obtained for $l + e = \dfrac{\lambda}{4}$

Sol.: $l + e = \frac{\lambda}{4}$ ...(i)

$l' + e = \frac{3\lambda}{4}$ ...(ii)

Solving we get, l' = 31 cm

24. Answer (3)

Hint: $v_{max} = A\omega$

Sol.: $v_{max} = A\omega$

= 4(2) = 8 m/s

25. Answer (3)

Hint: Minimum time taken is from $\left(-\frac{A}{2} \text{ to } +\frac{A}{2}\right)$

Maximum time taken from (0 – A)

Sol.: Time taken will be minimum when it travels from $\left(-\frac{A}{2} \text{ to } \frac{A}{2}\right)$

$T_{min} = \frac{T}{12} + \frac{T}{12} = \frac{T}{6} = 2$ s

$T_{max} = \frac{T}{4} = 3$ s

$\frac{T_{min}}{T_{max}} = \frac{2}{3}$

26. Answer (1)

Hint: Ideal gas possess kinetic energy only.

Sol.: Average kinetic energy $\left(\frac{1}{2}m\bar{v}^2\right) = \frac{3}{2}k_BT$

$v_{rms} = (\bar{v}^2)^{1/2} = \sqrt{\frac{3k_BT}{m}}$

27. Answer (3)

Hint: By Wien's law $\lambda_m \propto \frac{1}{T}$

Sol.: $\frac{(\lambda_m)_1}{(\lambda_m)_2} = \frac{T_2}{T_1}$

$\frac{\lambda_m}{(\lambda_m)_2} = \frac{3000}{2000}$

$(\lambda_m)_2 = \frac{2}{3}\lambda_m$

28. Answer (3)

Hint: Thermal capacity $H = ms$

Sol.: $H = ms$

Since material is same

$\therefore \frac{H_1}{H_2} = \frac{m_1}{m_2} = \frac{V_1}{V_2}$

$\frac{H_1}{H_2} = \left(\frac{R_1}{R_2}\right)^3 = \frac{1}{8}$

29. Answer (4)

Hint: Efficiency of Carnot engine, $\eta = 1 - \frac{T_L}{T_H}$

Sol.: $0.4 = 1 - \frac{T_2}{T}$, $T_2 = 0.6T$

$0.5 = 1 - \frac{T_2}{T'}$, $0.5 = \frac{0.6T}{T'}$, $T' = \frac{6}{5}T$

30. Answer (2)

Hint: Characteristics of standing waves

Sol.: Two travelling waves of same amplitude frequency and velocity moving in opposite directions are superposed, the phenomenon observed is standing waves.

31. Answer (4)

Hint: Average K.E $\propto$ Temperature

Sol.: $E = \frac{5}{2}kT$

For $\frac{E_{H_2}}{E_{O_2}} = \frac{\frac{5}{2}k\ 373}{\frac{5}{2}k\ 473}$

$E_{O_2} = 1.27E$

32. Answer (2)

Hint: $(C_V)_{mix} = \frac{n_1C_{V_1} + n_2C_{V_2}}{n_1 + n_2}$

Sol.: $C_V = \frac{f}{2}R$

$C_{V_1} = \frac{5}{2}R, \quad n_1 = 2$

$C_{V_2} = \frac{3}{2}R, \quad n_2 = 4$

$(C_V)_{mix} = \frac{2 \times \frac{5}{2}R + 4 \times \frac{3}{2}R}{6} = \frac{11}{6}R$

33. Answer (2)

Hint & Sol.: $v_{rms} = \sqrt{\frac{3RT}{M}}$, $P = \frac{1}{3}\rho(v_{rms})^2$,

$U = nC_vT$, Translational $K.E. = \frac{3}{2}K_BT$

34. Answer (2)

Hint: For polytropic process

$C = C_v + \frac{R}{1-N}$, PV^N = constant

Sol.: $C = C_V + \frac{R}{1-N}$

$R = \frac{3}{2}R + \frac{R}{1-N} \quad N = 3$

$\therefore$ PV^3 = constant

35. Answer (2)

Hint: $v_{\text{rms}} = \sqrt{\frac{3RT}{M}}$

Sol.: $v_{\text{rms}} \propto \sqrt{T}$

$\frac{v}{v'} = \frac{\sqrt{T}}{\sqrt{T'}}$

$T' = T + 3T = 4T$

$v' = 2v$

SECTION-B

36. Answer (1)

Hint & Sol.: Work done in adiabatic process is given by $W = \frac{nR(T_1 - T_2)}{\gamma - 1}$

$= \frac{P_1V_1 - P_2V_2}{\gamma - 1}$

37. Answer (2)

Hint: Total degree of freedom of a diatomic gas is 5.

Sol.: $C_v = \frac{R}{\gamma - 1}$

$\gamma = \frac{7}{5}$

$C_v = \frac{R}{\frac{7}{5} - 1} = \frac{5}{2}R$

38. Answer (1)

Hint & Sol.: Volume of the cavity will increase while heating.

39. Answer (3)

Hint & Sol.: Sound waves transfer both energy and momentum.

40. Answer (2)

Hint: $y = A\sin\omega t$

$v = \omega\sqrt{A^2 - y^2}$

Sol.: At $\frac{T}{12}$, $y = \frac{A}{2}$

$v = \omega\sqrt{A^2 - \frac{A^2}{4}} = \frac{\sqrt{3}A\omega}{2}$

41. Answer (1)

Hint: Time period of simple pendulum $T \propto \sqrt{\ell}$

Sol.: $T = 2\pi\sqrt{\frac{\ell}{g}} \Rightarrow T \propto \sqrt{\ell}$

$\frac{2}{T'} = \frac{1}{\sqrt{1 - \frac{7}{16}}} \qquad \frac{2}{T'}, = \frac{4}{3}$

$T' = \frac{3}{2} = 1.5$ second

42. Answer (4)

Hint: Use ideal gas equation.

Sol.: $PV = nRT$

$n = \frac{PV}{RT}$

$n = \frac{1}{2} = 16$ g of O_2 = 1 g of H_2

43. Answer (3)

Hint: Equation of travelling wave moving along negative x-axis is given by $y = A\sin(\omega t + kx)$

Sol.: $k = \frac{2\pi}{\lambda} \; \therefore k = \frac{2\pi}{0.5} = 4\pi$

$y = 0.25\sin(\omega t + 4x)$

44. Answer (2)

Hint: $\frac{dQ}{dt} = KA\frac{dT}{dx}$

Sol.: $\frac{\left(\frac{dQ}{dt}\right)_A}{\left(\frac{dQ}{dt}\right)_B} = \frac{K_1A_1\frac{dT}{l_1}}{K_2A_2\frac{dT}{l_2}}$

$\frac{4}{1} = \frac{K_1A_1}{K_2A_2}\,\frac{l_2}{l_1}$

$K_1A_1 = 8K_2A_2$

45. Answer (1)

Hint: Use $v = \omega\sqrt{A^2 - x^2}$

Sol.: $v = \omega\sqrt{A^2 - x^2}$

$= 2\sqrt{(6)^2 - (2)^2}$

$= 2\sqrt{32} = 2 \times 5.65 = 11.3$ cm/s

46. Answer (1)

Hint: Time period of spring pendulum is given by

$$T = 2\pi\sqrt{\frac{m}{k}}$$

Sol.: $T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{0.2}{80}} = 0.314\text{ s}$

47. Answer (3)

Hint & Sol.: Standing wave is given by $y = 2A\sin kx\cos\omega t$

48. Answer (1)

Hint: Use equation $\left(\frac{dQ}{dt}\right)_A + \left(\frac{dQ}{dt}\right)_B + \left(\frac{dQ}{dt}\right)_C = 0$

Sol.: $KA\frac{(70-T_0)}{l} + \frac{KA(70-T_0)}{l} + \frac{KA(40-T_0)}{l} = 0$

$$T_0 = \frac{180}{3} = 60°\text{C}$$

49. Answer (2)

Hint: In simple harmonic motion

Displacement $y = A\sin\omega t$

Acceleration $a = -\omega^2 y$

Velocity $v = \omega\sqrt{A^2 - y^2}$

Sol.: $v = \omega\sqrt{A^2 - y^2}$

$$v^2 = A^2\omega^2 - \omega^2 y^2$$

$$\frac{v^2}{A^2\omega^2} + \frac{y^2}{A^2} = 1$$

Equation of ellipse.

50. Answer (2)

Hint: In string fixed at both ends,

3rd overtone = 4(fundamental frequency)

Sol.: Third overtone is 4th harmonic

$4f = 600$

$f = 150$ Hz

[CHEMISTRY]

SECTION-A

51. Answer (1)

Hint and Sol: Decomposition of gaseous ammonia on a hot platinum surface at high pressure is an example of zero order reaction

$$2NH_3(g) \xrightarrow[\text{Pt catalyst}]{1130\text{ K}} N_2(g) + 3H_2(g)$$

52. Answer (3)

Hint: Nucleophiles are electron rich species.

Sol.: Nucleophiles attack at low electron density site and they are nucleus seeking species.

53. Answer (1)

Hint: % of N = $\frac{28\text{ x}}{22400 \times \text{W}} \times 100$

x is the volume of N_2 at STP.

Sol.: W = 0.30 g $\quad$ V_1 = 60 mL

P_1 = (715 – 15) = 700 mm of Hg

T_1 = 300 K

Now $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$

$$V_2 = \frac{P_1V_1}{T_1} \times \frac{T_2}{P_2} = \frac{700 \times 60 \times 273}{300 \times 760}$$

V_2 = 50.29 mL

% of N = $\frac{28 \times 50.29}{22400 \times 0.30} \times 100$

= 20.95%

54. Answer (3)

Hint: Structural isomers have same molecular formula but different structure.

Sol.: Possible structural isomers of $C_4H_{10}O$ are

$CH_3 - CH_2 - CH_2 - CH_2 - OH$

```
        H
        |
CH3 – C – CH2 – OH
        |
        CH3
```

```
CH3 – CH2 – CH – CH3
                |
                OH
```

```
        OH
        |
CH3 – C – CH3
        |
        CH3
```

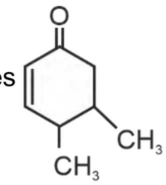
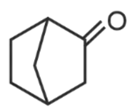
```
CH3 – CH – O – CH3
        |
        CH3
```

$CH_3 - CH_2 - O - CH_2 - CH_3$

$CH_3 - CH_2 - CH_2 - O - CH_3$

55. Answer (1)

Hint: If a compound have enoliable hydrogen then it will show keto-enol Tautomerism.

Sol.: The molecules  and 

have α – Hydrogen which can involve in Tautomerism.

56. Answer (1)

Hint: Kjeldahl's method is not applicable to compounds containing nitrogen in nitro and azo groups and nitrogen present in the ring (e.g Pyridine)

Sol.: Kjeldahl's method is applicable for amine ($CH_3 – CH_2 – NH_2$).

57. Answer (1)

Hint: In resonance electrons are delocalised.

Sol.: $\overset{\ominus}{C}H_2 - \overset{O}{\overset{\|}{C}} - CH_2 - CH_3 \leftrightarrow CH_2 = \overset{:\ddot{O}:^{\ominus}}{\overset{|}{C}} - CH_2 - CH_3$ are resonating structures.

58. Answer (4)

Hint: Polarization of σ bonded electron due to difference in electronegativity is known as inductive effect.

Sol.: Correct order of – I effect.

– CN > – F > – OH

59. Answer (2)

Hint and Sol.:

	Column I (Compound)		**Column II (Common Name)**
a.	$C_6H_5OCH_3$	(i)	Anisole
b.	$C_6H_5COCH_3$	(ii)	Acetophenone
c.	$CH_3OCH_2CH_3$	(iii)	Ethyl methyl ether
d.	$C_6H_5NH_2$	(iv)	Aniline

60. Answer (3)

Hint: Nitrogen, sulphur, halogens and phosphorus present in an organic compound are detected by "Lassaigne's test".

Sol.: Carbon and hydrogen are detected by heating the compounds with cupric oxide.

61. Answer (4)

Hint: Steam distillation is applied to separate substances which are steam volatile and are immiscible with water.

Sol.: Water and Aniline is separated by steam distillation method.

62. Answer (2)

Hint and Sol: Thin layer chromatography is a type of adsorption chromatography where as paper chromatography is type of partition chromatography.

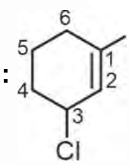
63. Answer (2)

Hint: Carbocations are stabilized by +R, hyperconjugation and +I effect.

Sol.: Order of stability of carbocation 3° > 2° > 1°

64. Answer (1)

Hint: Double bond have more priority than substituents in IUPAC naming.

Sol.:  3-Chloro-1-methylcyclohex-1-ene.

65. Answer (2)

Hint: $CH \equiv C - CH_2 - \underset{CH_3}{\underset{|}{C}} = CH_2$

Sol.: Number of sp^2 hybridised carbon atoms = 2

Number of π – bonds = 3

66. Answer (3)

Hint: Rate = $PZ_{AB}\ e^{-E_a/RT}$

Sol.: $e^{-E_a/RT}$ represents the fraction of molecules with energies equal to or greater than E_a.

67. Answer (1)

Hint: $k = Ae^{-E_a/RT}$

Sol.: $\ln k = \ln A - \dfrac{E_a}{RT}$

Slope = $\dfrac{-E_a}{R}$, Intercept = ln A

$$\frac{\text{Slope}}{\text{Intercept}} = \frac{-E_a/R}{\ln A} = -\frac{E_a}{R \times \ln A}$$

68. Answer (3)

Hint: Rate = $k[A]^x[B]^y$ where x and y are the order w.r.t. A and B

Sol.: $r = k[A]^x[B]^y$ …(i)

$2r = k[2A]^x[B]^y$ …(ii)

$8r = k[2A]^x[2B]^y$ …(iii)

From equation (i), (ii) and (iii) we get

x = 1, y = 2

Rate = $k[A]^1[B]^2$

69. Answer (2)

Hint and Sol.: Order of reaction is applicable to elementary as well as complex reactions.

70. Answer (3)

Hint: $t = \frac{2.303}{k}\log\frac{[A]_0}{[A]_t}$

Sol.: $t = \frac{2.303}{k}\log\frac{100}{100-99.99}$

$= \frac{2.303}{k}\log\frac{100}{0.01}$

$= \frac{2.303}{k}\log 10^4$

$= \frac{4\times 2.303}{k}$

71. Answer (1)

Hint: $r = \frac{-d\left[BrO_3^-\right]}{dt} = \frac{-1}{5}\frac{d[Br^-]}{dt} = -\frac{1}{6}\frac{d[H^+]}{dt}$

$= \frac{1}{3}\frac{d[Br_2]}{dt} = \frac{1}{3}\frac{d[H_2O]}{dt}$

Sol.: $\frac{\text{Rate of appearance of } H_2O}{\text{Rate of disappearance of } Br^-} = \frac{3r}{5r} = \frac{3}{5}$

72. Answer (1)

Hint: $t_{1/2} \propto a^{1-n}$

Sol.: For zero order reaction

$t_{1/2} \propto a^{1-0} \propto a^1$

Means half-life of zero order reaction is directly proportional to initial concentration of the reactant.

73. Answer (2)

Hint: Slowest step is the rate determining step

Sol.:

(1) $A_3 \underset{}{\overset{\text{Fast}}{\rightleftharpoons}} A_2 + A$

$K_{eq} = \frac{[A_2][A]}{[A_3]}$

$[A] = K_{eq} \times \frac{[A_3]}{[A_2]} \quad \ldots(1)$

(2) $A_3 + A \xrightarrow{\text{slow}} 2A_2$

$r = K[A_3][A]$

$= K[A_3]K_{eq}\frac{[A_3]}{[A_2]}$

$= K'[A_3]^2[A_2]^{-1}$

Overall order = (2) + (–1) = 1

74. Answer (3)

Sol.:

	$A(g)$	$\rightarrow$	$2B(g)$	+	$C(g)$
t = 0	0.1				
t = 115	0.1 – x		2x		x

0.1 + 2x = 0.28

2x = 0.18

x = 0.09

$k = \frac{2.303}{115}\log\left(\frac{0.1}{0.1-0.09}\right)$

$k = 2 \times 10^{-2}\ sec^{-1}$

75. Answer (3)

Hint: π = iCRT

i = 1

Sol.: $2.57\times10^{-3} = \frac{\frac{2.52}{M}}{\frac{400}{1000}}\times0.083\times300$

$2.57\times10^{-3} = \frac{2.52\times1000}{M\times400}\times0.083\times300$

$M = \frac{2.52\times10\times0.083\times300}{2.57\times4\times10^{-3}}$

$= \frac{2.52\times0.083\times3\times10^6}{2.57\times4}$

= 61038 g mol^{-1}

76. Answer (4)

Hint: $X_A = \frac{n_A}{n_A+n_B}$

Sol.: 2 m aqueous solution means that 2 mole solute present in 1 kg of water.

$X = \frac{2}{2+\frac{1000}{18}} = \frac{2}{2+55.5} = \frac{2}{57.5} = 0.035$

77. Answer (4)

Hint: Value of K_f (molal depression constant) depends on the nature of solvent.

Sol.: $K_f = \frac{RT_f^2}{1000\ \Delta H}$

ΔH represents enthalpy of fusion.

So, K_f is independent of molality of the solution.

78. Answer (2)

Hint: Azeotropes are binary mixture having the same composition in liquid and vapour phase and boil at a constant temperature.

Sol.: Nitric acid and water can form maximum boiling azeotrope.

79. Answer (4)

Hint: $\Delta T_b = i\,K_b\,m$

Sol.:

(1) 0.1 m KCl (aq)

$i = 2$

$\Delta T_b = 2 \times 0.1\,K_b = 0.2\,K_b$

(2) 0.20 m $C_6H_{12}O_6$ (aq)

$i = 1$

$\Delta T_b = 1 \times 0.2 \times K_b = 0.2\,K_b$

(3) 0.10 m $Al_2(SO_4)_3$ (aq)

$i = 5$

$\Delta T_b = 5 \times 0.10 \times K_b = 0.5\,K_b$

(4) 0.20 m K_2SO_4 (aq)

$i = 3$

$\Delta T_b = 3 \times 0.2 \times K_b = 0.6\,K_b$

80. Answer (4)

Hint: Amount of dissolved oxygen is more in cold water than in warm water.

Sol.: Due to greater oxygen content in cold water, aquatic species are more comfortable in cold water than warm water. Solubility of gases in liquid increases with decrease in temperature.

81. Answer (2)

Hint: $P = K_H X$

Sol.: Higher the volume of K_H at a given temperature and pressure, the lower is the solubility of the gas in the liquid.

82. Answer (3)

Hint: Volume changes with change in temperature.

Sol.: Molality and % (w/w) terms are not related with volume therefore these are temperature independent.

83. Answer (2)

Hint: $y_A = \dfrac{P_A^o X_A}{P_A^o X_A + P_B^o X_B}$

Sol.: $P_A = P_A^o X_A = 250$

$$X_A = \frac{250}{500} = \frac{1}{2}$$

$$X_B = 1 - \frac{1}{2} = \frac{1}{2}$$

$$y_A = \frac{P_A^o X_A}{P_A^o X_A + P_B^o X_B}$$

$$= \frac{250}{250 + 200 \times \frac{1}{2}}$$

$$y_A = \frac{250}{250+100} = \frac{250}{350} = \frac{25}{35} = \frac{5}{7}$$

$$y_B = 1 - \frac{5}{7} = \frac{2}{7}$$

$$\text{Molar ratio} = \frac{y_A}{y_B} = \frac{5/7}{2/7} = \frac{5}{2}$$

84. Answer (3)

Hint: For negative deviation from Raoult's law, the interactions between A---A, B---B weaker than A---B.

Sol.: Due to intermolecular H-bonding, the solution of acetone and chloroform show negative deviation from Raoult's Law.

85. Answer (2)

Hint: $\Delta T_f = i\,K_f\,m$

Sol.: For P_1 solution;

$$im = i \times \frac{20}{180} \times \frac{1000}{500} = 0.22\text{ m}$$

$(\Delta T_f)_1 = 0.22\,K_f$

$(T_f)_1 = -\,0.22\,K_f$

For P_2 solution, $im = 1 \times \dfrac{20}{60} \times \dfrac{1000}{500}$

$= 0.66$ m

$(\Delta T_f)_2 = 0.66\,K_f$

$(T_f)_2 = -\,0.66\,K_f$

For P_3 solution, $im = 1 \times \dfrac{20}{342} \times \dfrac{1000}{500}$

$= 0.12$ m

$(\Delta T_f)_3 = 0.12\,K_f$

$(T_f)_3 = -\,0.12\,K_f$

Decreasing order of freezing point of given solutions is $P_3 > P_1 > P_2$.

SECTION-B

86. Answer (1)

Hint and Sol: Catalyst catalyse the spontaneous reactions but does not catalyses non-spontaneous reactions.

87. Answer (2)

Hint: $\log\frac{K_2}{K_1} = \frac{E_a}{2.303\,R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$

Sol.: $\log\frac{K_2}{K_1} = \frac{14000}{2.303\,R}\left[\frac{1}{400} - \frac{1}{500}\right]$

$\log\frac{K_2}{K_1} = 0.365$, so $\frac{K_2}{K_1} = 2.32$

$K_2 = 3.7 \times 10^8$

88. Answer (2)

Hint and Sol: If nitrogen or sulphur is present in the compound, the sodium fusion extract is first boiled with concentrated nitric acid to decompose cyanide or sulphide of sodium formed during Lassaigne's test.

89. Answer (2)

Hint: Percentage of halogen (X)

$$= \frac{\text{atomic mass of } X \times m_1 \times 100}{\text{molecular mass of } AgX \times m}$$

m_1 = mass of AgX

m = mass of organic compound

Sol.: m_1 = 1.128 g, m = 1 g

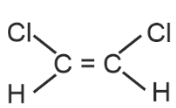
M(AgBr) = 188 g mol^{-1}

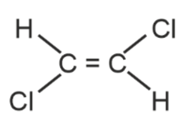
% of Br = $\frac{80 \times 1.128 \times 100}{188 \times 1} = 48\%$

90. Answer (4)

Hint: If the two atoms or groups attached to each *sp*^{2} hybridised carbon atom of alkene are different, they can show geometrical isomerism.

Sol.: Following molecules are two geometrical isomer

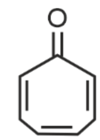

cis form


trans form

91. Answer (4)

Hint: Aromatic compounds containing benzene ring are known as benzenoids

Sol.:


Tropone

Non-benzenoid aromatic compound

92. Answer (4)

Hint and Sol: Correct order of stability of free radical II > III > I

$CH_2 = CH - \dot{C}H_2 \longleftrightarrow \dot{C}H_2 - CH = CH_2$

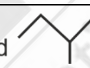
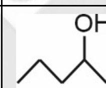
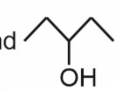
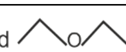
[two equivalent resonating structure]

Therefore, allylic radical is more stable than benzylic radical.

93. Answer (1)

Hint: Structural isomers have same molecular formula but different structure.

Sol.:

	Column I (Structures)	**Column II (Type of isomers)**
a.	 and 	Chain isomers
b.	 and 	Position isomers
c.	$CH_3 - CH_2 - CHO$ and $CH_3 - CO - CH_3$	Functional group isomers
d.	 and 	Metamers

94. Answer (3)

Hint: $\frac{r_{(t+\Delta t)}}{r_t} = (2)^{\frac{\Delta t}{10}}$

Sol.: $\frac{r_{25}}{r_{55}} = \frac{1}{2^{\frac{55-25}{10}}} = \frac{1}{2^3} = \frac{1}{8}$

95. Answer (1)

Hint: Relative lowering in vapour pressure = mole fraction of solute.

Sol.: $\frac{\Delta P}{P_A^o} \times 100 = 40$

$x_B = 0.4$

$$x_B = \frac{n_B}{n_A + n_B} = \frac{w/60}{\frac{171}{114} + \frac{w}{60}} = 0.4$$

$$\frac{\frac{w}{60}}{\frac{171}{114}+\frac{w}{60}}=\frac{4}{10}=\frac{2}{5}$$

$$\frac{5w}{60}=2\times\frac{171}{114}+\frac{2w}{60}$$

$$\frac{3w}{60}=2\times\frac{171}{114}$$

$$w=2\times20\times\frac{171}{114}=60$$

96. Answer (2)

Hint: $i = 1 - \alpha\left(1-\frac{1}{n}\right)$

Sol.: $1 - 0.3\left(1-\frac{1}{3}\right)$

$= 1 - 0.3 + 0.1 = 0.80$

97. Answer (3)

Hint: $\Delta T_b = i\,K_b m$, $\Delta T_f = i\,K_f m$

Sol.: i = 1 for urea

$\Delta T_b = 101.024 - 100 = 1.024°C$

$1.024 = 0.512 \times m$

$m = 2$

$\Delta T_f = 1.86 \times 2 = 3.72°C$

$0 - T_f = 3.72°C$

$T_f = -3.72°C$

98. Answer (4)

Hint: According to IUPAC nomenclature carboxylic acid and its derivatives are at higher priority.

Sol.: Correct order of priority of functional groups in IUPAC naming is

$-COOH > -COCl > -OH > -NH_2$

99. Answer (3)

Hint: For isotonic solution

$$\boxed{\pi_1 = \pi_2}$$

Sol.: $\pi_1 = \pi_2$

$C_1RT = C_2RT$

$C_1 = C_2$

$$\frac{6.84\times10}{342}=\frac{2\times10}{M}$$

$$2\times10^{-2}=\frac{2}{M}$$

$$M=\frac{1}{10^{-2}}=100\ \text{g/mol}$$

100. Answer (2)

Hint and Sol.: Osmotic pressure is widely used to determine molar masses of proteins and polymers because pressure measured is around the room temperature and the molarity of solution is used instead of molality.

[BOTANY]

SECTION-A

101. Answer (4)

Hint: Appearance of different forms of leaves on the same plant is called heterophylly.

Sol.: Buttercup shows environmental heterophylly.

102. Answer (2)

Hint: The enzyme is attached to the inner membrane of mitochondria.

Sol.: Succinate dehydrogenase is the enzyme which is a part of both Krebs' cycle and ETS.

103. Answer (3)

Hint: This phytohormone is the only gaseous hormone.

Sol.: Cousins discovered that ripened oranges emitted a volatile substance, which caused early ripening of bananas kept nearby. Later on, this volatile substance was identified as ethylene.

104. Answer (4)

Hint: Apical dominance is caused by auxins.

Sol.: Cytokinins help in the production of new leaves and chloroplast in leaves.

105. Answer (2)

Hint: This hormone, along with auxins is responsible for callus formation.

Sol.: Cytokinins and auxins are involved in morphogenesis or organogenesis.

106. Answer (4)

Hint: The chief photosynthetic pigment in green plants is chlorophyll *a*.

Sol.: Reaction centre in P_{700} and P_{680} is formed by chlorophyll *a*.

107. Answer (3)

Hint: The enzyme is involved in the synthesis of phosphoenol pyruvate.

Sol.: PEP synthetase is the cold sensitive enzyme of the C_4 pathway.

108. Answer (4)

Hint: Removal of apical bud promotes the growth of lateral buds.

Sol.: Application of auxins, like IBA, to the cut portion of decapitated shoots, prevents the growth of side branches.

109. Answer (1)

Hint: Substrate-level phosphorylation refers to the direct formation of ATP or GTP by transferring a phosphate group from a high energy compound to an ADP or GDP molecule.

Sol.: In each turn of citric acid cycle, there is a single substrate level phosphorylation. For glucose molecule there will be occurrence of 2 substrate level phosphorylation reactions during the citric acid cycle.

110. Answer (3)

Hint: Granum refers to the stack of thylakoids.

Sol.: Pigments that help in light reactions are embedded in the thylakoid membrane.

111. Answer (2)

Hint: Redifferentiated cells lose their ability to divide.

Sol.: Secondary xylem and cork are the examples of redifferentiated tissues.

112. Answer (2)

Hint: The first 6-carbon compound is formed by the condensation of acetyl CoA with oxaloacetic acid and H_2O.

Sol.: The first step of Krebs' cycle involves the synthesis of citric acid. This reaction is catalysed by citrate synthase.

113. Answer (3)

Hint: In cyclic photophosphorylation only ATP is synthesized.

Sol.: In cyclic photophosphorylation P_{700} is the reaction centre.

114. Answer (4)

Hint: Gibberellins stimulate the synthesis of hydrolytic enzymes, which helps in seed germination.

Sol.: ABA (Abscisic acid) inhibits gibberellin mediated amylase formation during germination of seed.

115. Answer (3)

Hint: Increase in the number of female flowers in cucumber is done by ethylene.

Sol.: Ethylene helps in initiating germination of seeds, such as peanuts.

116. Answer (2)

Hint: Chlorophyll *a* is the chief photosynthetic pigment.

Sol.: Since the action spectrum of photosynthesis corresponds closely to the absorption spectrum of chlorophyll *a*, it helped in concluding that chlorophyll *a* is the chief photosynthetic pigment.

117. Answer (1)

Hint: When fats and proteins are the respiratory substrate, the value of RQ is less than one.

Sol.: The RQ of fatty acid, tripalmitin is 0.7.

118. Answer (2)

Hint: There is an equal number of utilization of ATP and NADPH in the reduction phase of C_3 cycle.

Sol.: There will be a requirement of 4 ATP and 4 NADPH in the reduction phase of Calvin cycle for two molecules of CO_2 fixed. The ratio obtained will be 1 : 1.

119. Answer (3)

Hint: Exponential growth shows a sigmoid growth curve.

Sol.: $W_1 = W_0e^{rt}$ is the correct equation for exponential growth.

120. Answer (1)

Sol.: The light saturation occurs at 10% of the total sunlight available to the plants.

121. Answer (2)

Hint: Splitting of water is associated with non-cyclic photophosphorylation.

Sol.: (i) Water is the electron donor in non-cyclic photophosphorylation.

(ii) Both PS II and PS I are the parts of non-cyclic photophosphorylation.

P_{680} is associated with PS II.

122. Answer (2)

Hint: Bakanae disease (foolish seedling disease) in rice was caused by a fungal pathogen, *Gibberella fujikuroi*.

Sol.: Gibberellin was isolated from the fungus *Gibberella fujikuroi*.

123. Answer (3)

Hint: Citrate synthase is the enzyme of TCA cycle.

Sol.: (i) Aldolase in an enzyme of the glycolytic pathway.

(ii) ATP synthase is involved in oxidative phosphorylation.

(iii) Pyruvate dehydrogenase is an enzyme of link reaction.

124. Answer (2)

Hint: During decarboxylation, CO_2 molecule is released.

Sol.: Decarboxylation does not occur during glycolysis and lactic acid fermentation.

125. Answer (3)

Hint: RuBP is the primary CO_2 acceptor in C_3 plants.

Sol.: The primary CO_2 acceptor in C_4 plants is phosphoenol pyruvate.

126. Answer (3)

Hint: There is one step in glycolysis where NADH + H^+ is formed from NAD^+; this is when 3-phosphoglyceraldehyde (PGAL) is converted to 1,3-bisphosphoglycerate.

Sol.: Total ATP produced from glycolysis:

2 NADH = 2 × 3 ATP	= 6 ATP
4 ATP	= 4 ATP
	= 10 ATP (Glycolysis)

Total ATP produced from link reaction and Krebs' cycle

8 NADH = 3 × 8	= 24 ATP
2 $FADH_2$ = 2 × 2	= 4 ATP
2 ATP	= 2 ATP
	= 30 ATP
So, Total ATP	= 40 ATP

127. Answer (3)

Hint: Joseph Priestley concluded that the plants restore to the air whatever the breathing mouse and the burning candle remove.

Sol.: (i) Jan Ingenhousz showed oxygen was released in presence of sunlight by green parts of plants.

(ii) Julius von Sachs proved that glucose is produced during photosynthesis in the green parts of the plant.

(iii) Cornelius van Niel showed that oxygen evolved by green plants comes from water and not from carbon dioxide.

128. Answer (1)

Hint: Intrinsic factors work from inside the cells.

Sol.: Genetic factors and PGRs are the most important intrinsic factors controlling the developmental processes of the plant.

129. Answer (2)

Hint: Redox equivalents in form of hydrogen atoms are removed and transferred to NAD^+.

Sol.: Two redox-equivalents are removed from one PGAL molecule during the conversion of PGAL to BPGA.

130. Answer (2)

Hint: Microorganisms show a typical geometric growth curve.

Sol.: Sigmoid growth curve is seen in the microorganisms, when they have limited food and space.

131. Answer (2)

Hint: Blackman gave the law of limiting factors in 1905.

Sol.: The first action spectrum of photosynthesis was described through the experiments performed by T.W. Engelmann.

132. Answer (4)

Hint: S-shaped curve is observed in geometric growth.

Sol.: An elongating root at constant rate shows arithmetic growth.

Geometric growth is seen when a zygote is developed into embryo.

133. Answer (3)

Hint: Floating respiration uses fat or carbohydrate as respiratory substrates.

Sol.: Protein is used as a substrate in protoplasmic respiration.

134. Answer (4)

Hint: Meristematic cells have a large conspicuous nucleus.

Sol.: Meristematic cells have a primary cell wall but large vacuole and secondary cell wall is absent.

135. Answer (4)

Hint: Increase in girth is caused by lateral meristems.

Sol.: Cork cambium is an example of secondary meristem responsible for secondary growth.

SECTION-B

136. Answer (4)

Hint: ABA is known as the stress hormone.

Sol.: Inhibitor B, abscission II and dormin were chemically similar inhibitors, collectively called ABA.

137. Answer (2)

Hint: ABA inhibits protein and RNA synthesis, so as to promote senescence of leaves.

Sol.: The first natural cytokinin was isolated from unripe maize grain and it was called zeatin.

138. Answer (2)

Hint: Assimilation number is the amount of CO_2 (in gms) assimilated by 1 g of chlorophyll in an hour.

Sol.: Assimilation number shows a relationship between the chlorophyll and photosynthesis.

139. Answer (1)

Hint: Charles Darwin and his son concluded that tip of coleoptile was the site of transmittable influence that caused the bending of the entire coleoptile.

Sol.: F.W. Went, isolated auxins from the tips of coleoptile of oat seedlings.

140. Answer (3)

Hint: The wall of bundle sheath cells are impervious to gaseous exchange.

Sol.: Bundle sheath cells of C_4 plants are characterized by large number of chloroplast, thick walls impervious to gaseous exchange and no intercellular spaces.

141. Answer (2)

Hint: Tryptophan acts as a precursor for auxins.

Sol.:

a. Acetyl CoA serves as a precursor for gibberellins.
b. Methionine is a precursor for ethylene.
c. Violaxanthin is precursor for forming abscisic acid.

142. Answer (2)

Hint: Methionine is the precursor of this plant hormone.

Sol.: Ethylene is the PGR which elongates stem in deep water rice plant.

143. Answer (2)

Hint: Abscission is shedding of leaves and fruits.

Sol.: Both ethylene and ABA promote shedding of leaves.

144. Answer (4)

Hint: Double carboxylation is a characteristic feature of C_4 plants.

Sol.: Bell pepper is a C_3 plant.

145. Answer (4)

Hint: Respiratory quotient value for proteins is less than unity.

Sol.: Respiratory quotient (RQ) of organic acids like oxalic acid is more than unity.

146. Answer (1)

Hint: Chlorophyll *b* shows maximum absorption in blue region of light spectrum.

Sol.: Blue region of light spectrum lies in range of 400-500 nm wavelength. In this region Chlorophyll *b* shows maximum absorption.

147. Answer (1)

Hint: The electrons needed to replace those removed from PS I are provided by photosystem II.

Sol.: Splitting of water molecule is mainly responsible for the source of electrons, used in replacing the ones lost from the reaction centre of PS II.

148. Answer (3)

Hint: Gibberellins cause fruits like apple to elongate.

Sol.: Ethylene is a gaseous hormone, which is used to synchronize fruit-set in pineapple and thining of fruits like cotton and cherry.

149. Answer (2)

Hint & Sol.: No other alternative substrates enter the pathway according to assumptions made for respiratory balance sheet.

150. Answer (2)

Hint: Substrate level phosphorylation is the direct formation of ATP, when phosphate is transferred from the substrate.

Sol.: There will be 2 substrate level phosphorylation reactions, one during glycolysis and one during Krebs' cycle when PEP is used as a substrate.

[ZOOLOGY]

SECTION-A

151. Answer (1)

Hint: Organism in which tracheoles are present

Sol.: Proboscis gland is an excretory structure in *Balanoglossus*.

Protonephridia/flame cells are the excretory structure in platyhelminths. *Anopheles* belongs to the phylum Arthropoda and has Malpighian tubules as the excretory structures.

152. Answer (3)

Hint: Aschelminths

Sol.: Female roundworms are longer than males. Tapeworm, liver fluke and earthworm are monoecious and do not show sexual dimorphism.

153. Answer (1)

Hint: Ventral side is the lower side in *Asterias*

Sol.: In echinoderms (*e.g.*, *Asterias*), digestive system is complete with mouth on the lower (ventral) side and anus on the upper (dorsal) side.

154. Answer (3)

Hint: Mantle cavity

Sol.: In phylum Mollusca, the body of animals is covered by a calcareous shell and is unsegmented with a distinct head, muscular foot and visceral hump. A soft and spongy layer of skin forms a mantle over the visceral hump. The space between the hump and the mantle is called mantle cavity in which feather-like gills are present.

The anterior head region has sensory tentacles. The mouth contains a file-like rasping organ for feeding called radula.

155. Answer (1)

Hint: Oxytocin

Sol.: The adrenal cortex secretes many hormones, commonly called corticoids. The corticoids, which are involved in carbohydrate metabolism are called glucocorticoids. Glucocorticoids stimulate gluconeogenesis, lipolysis and proteolysis and inhibit cellular uptake and utilisation of amino acids.

The α-cells of islets of Langerhans secrete a hormone called glucagon, while the β-cells secrete insulin. Glucagon is a peptide hormone, and plays an important role in maintaining the normal blood glucose levels. Glucagon acts mainly on the liver cells (hepatocytes) and stimulates glycogenolysis resulting in an increased blood sugar.

156. Answer (2)

Hint: Also known as sea walnuts

Sol.: Ctenophores are exclusively marine. Many organisms in the phylum Platyhelminthes, Annelida and Mollusca are freshwater.

157. Answer (1)

Hint: Cause for osteoporosis in females

Sol.: Decrease in the oestrogen levels can lead to increased chances of fracture in old age in females.

Low pitch is seen in males as compared to females. Testosterone affects the male libido and maturation of epididymis. Estrogen is secreted from growing follicles and corpus luteum in females. Glucocorticoids, particularly cortisol, produces anti-inflammatory reactions and suppresses the immune response.

158. Answer (3)

Hint: Acts on the thyroid follicles

Sol.: In humans, each thyroid follicle is composed of follicular cells, enclosing a cavity. These follicular cells synthesize two hormones, tetraiodothyronine or thyroxine (T4) and triiodothyronine (T3).

Iodine is essential for the normal rate of hormone synthesis in the thyroid gland. Deficiency of iodine in our diet results in hypothyroidism and enlargement of the thyroid gland is called goitre. During goitre, TSH levels increase which stimulates thyroid gland to synthesize more thyroxine.

TSH stimulates the synthesis and secretion of thyroid hormones from the thyroid gland.

159. Answer (2)

Hint: *Hemidactylus* is a reptile.

Sol.:

Scientific name	**Common name**
Antedon	Sea lily
Dentalium	Tusk shell
Ornithorhynchus	Platypus
Hemidactylus	Wall lizard

160. Answer (4)

Hint: Possesses three – chambered heart

Sol.: Arthropods, hemichordates and molluscs (except cephalopods) possess an open circulatory system. Annelids (except leech) and chordates have closed circulatory system.

Periplaneta americana and *Locusta* – Arthropods

Saccoglossus – Hemichordate

Rana – Chordate

161. Answer (2)

Hint: Vasa efferentia opens into Bidder's canal

Sol.: In frog, male reproductive organs consist of a pair of yellowish ovoid testes, which are found adhered to the upper part of the kidneys by mesorchium.

Vasa efferentia are 10-12 in number that arise from testes. They enter the kidneys on their side and open into Bidder's canal. Finally, it communicates with the urinogenital duct that comes out of the kidneys and open into the cloaca. Here, the ureters act as urinogenital duct. Urinary bladder is present ventral to the rectum which also opens in the cloaca.

The cloaca is a small, median chamber that is used to pass faecal matter, urine and sperms to the exterior.

Fat bodies do not open into cloaca.

162. Answer (3)

Hint: Exclude the specialised venous connections

Sol.: In frogs, partially digested food called chyme is passed from stomach to the first part of the small intestine, the duodenum. The duodenum receives bile from gall bladder and pancreatic juices from the pancreas through a common bile duct.

Special venous connection between liver and intestine as well as the kidney and lower parts of the body are present in frogs. The former is termed as hepatic portal system and latter is called renal portal system.

163. Answer (3)

Hint: Discharged during copulation

Sol.: In male cockroach, the sperms are stored in the seminal vesicles and are glued together in the form of bundles called spermatophores. On an average, female cockroach produces 9-10 oothecae, each containing 14-16 eggs. The development of *P. americana* is paurometabolous, meaning there is development through nymphal stage. The nymph looks very much like adults. The next to last nymphal stage has wing pads but only adult cockroaches have wings.

In cockroach, three ganglia lie in the thorax and six in the abdomen.

164. Answer (4)

Hint: Present at the junction of midgut and hindgut

Sol.: In cockroach, gizzard or proventriculus helps in grinding of the food particles. A ring of 6-8 blind tubules called hepatic or gastric caeca is present at the junction of foregut and midgut, which secrete digestive juice.

At the junction of midgut and hindgut, is present another ring of 100-150 yellow coloured thin filaments called Malpighian tubules.

165. Answer (1)

Hint: Thyroxine is an iodothyronine.

Sol.: Hormones which interact with intracellular receptors (mostly nuclear receptors), regulate gene expression or chromosome function by the interaction of hormone-receptor complex with the genome.

E.g., steroid hormones, iodothyronines, *etc*.

Steroid hormones – (Cortisol, testosterone, estradiol and progesterone).

Iodothyronine – (Thyroxine).

166. Answer (2)

Hint: Inhibitory hormone released from hypothalamus

Sol.: The hormones produced by hypothalamus are of two types, the releasing hormones (which stimulate secretion of pituitary hormones) and the inhibiting hormones (which inhibit secretions of pituitary hormones). GnRH stimulates the pituitary synthesis of gonadotrophins.

On the other hand, somatostatin from the hypothalamus inhibits the release of growth hormone from the pituitary. PRL and FSH are released by anterior pituitary.

167. Answer (2)

Hint: *Obelia* shows alternation of generation

Sol.: Bioluminescence is a well-marked property in ctenophores.

Few cnidarians exhibit metagenesis (alternation of generation) *i.e.*, polyps produce medusae asexually and medusae form the polyps sexually (*e.g.*, *Obelia*).

In molluscs, body is covered by a calcareous shell and is unsegmented with a distinct head, muscular foot and visceral hump.

168. Answer (1)

Hint: Capillaries are present in closed circulatory system.

Sol.: Cyclostomes have a sucking and circular mouth without jaws. They have an elongated body bearing 6-15 pairs of gill slits for respiration.

Circulation is of closed type. Cyclostomes are marine but migrate for spawning to freshwater. After spawning, within a few days, they die. Their larvae, after metamorphosis return to the ocean.

169. Answer (2)

Hint: Present between forebrain and hindbrain

Sol.: In frog, the brain is divided into forebrain, midbrain and hindbrain. Forebrain includes olfactory lobes, paired cerebral hemispheres and an unpaired diencephalon.

The midbrain is characterized by a pair of optic lobes. Hindbrain consists of cerebellum and medulla oblongata.

The medulla oblongata passes out through the foramen magnum and continues into spinal cord, which is enclosed in the vertebral column.

170. Answer (4)

Hint: Organ involved in hearing.

Sol.: Frog has different types of sense organs, namely organs of touch (sensory papillae), taste (taste-buds), smell (nasal epithelium), vision (eyes) and hearing (tympanum with internal ears). Out of these, eyes and internal ears are well-organised structures and the rest are cellular aggregations around nerve endings.

171. Answer (1)

Hint: Absence of air bladder

Sol.:

	Chondrichthyes	**Osteichthyes**
(1)	Skin is tough, containing minute placoid scales.	Skin is covered with cycloid/ctenoid scales.
(2)	Gill slits are without operculum.	They have four pairs of gills which are covered by an operculum on each side.
(3)	Due to absence of air bladder, they have to swim constantly to avoid sinking.	Air bladder is present which regulates buoyancy.

172. Answer (3)

Hint: Cephalochordate

Sol.: In the members of subphylum Cephalochordata and class Chondrichthyes, notochord is persistent throughout life.

Cephalochordate – *Branchiostoma*

Cyclostomes – *Petromyzon*, *Myxine*

Chondrichthyes – *Scoliodon*, *Pristis*, *Carcharodon*

In urochordates (*e.g.*, *Ascidia*, *Doliolum*, *Salpa*) notochord is present only in larval tail.

173. Answer (2)

Hint: Associated with glycosuria

Sol.: Diabetes mellitus – Associated with loss of glucose through urine and formation of harmful compounds known as ketone bodies. It could be caused due to prolonged hyperglycemia.

Acromegaly – Caused due to excess secretion of growth hormone in adults especially in middle age.

Addison's disease – Caused due to underproduction of hormones secreted by the adrenal cortex.

Exopthalmic goitre – A form of hyperthyroidism, also known as Graves' disease.

174. Answer (2)

Hint: Peptide hormones

Sol.: Hormones which interact with membrane bound receptors normally do not enter the target cell, but generate second messengers (*e.g.*, cyclic AMP, IP_3, Ca^{++}, *etc.*) which in turn regulate cellular metabolism.

Peptide, polypeptide and protein hormones generate second messengers. *E.g.*, Insulin, glucagon, pituitary hormones, hypothalamic hormones, *etc.*,

Also, endocrine cells present in different parts of the gastro-intestinal tract secrete four major peptide hormones, namely gastrin, secretin, CCK and GIP.

Estradiol and testosterone are steroid hormones.

175. Answer (1)

Hint: PTH is a hypercalcemic hormone.

Sol.: The parathyroid glands secrete a peptide hormone called parathyroid hormone (PTH). The secretion of PTH is regulated by the circulating levels of calcium ions. PTH acts on the bones and stimulates the process of bone resorption. Hypersecretion of PTH increases the bone resorption and can be a possible cause of weakening of bones.

176. Answer (3)

Hint: One of them causes uterine contraction

Sol.: Neurohypophysis (pars nervosa) also known as posterior pituitary, stores and releases two hormones called oxytocin and vasopressin, which are actually synthesized by the hypothalamus and are transported axonally to neurohypophysis.

PRL, TSH, ACTH, LH are released by anterior pituitary.

GnRH and somatostatin are released by hypothalamus.

177. Answer (1)

Hint: Body cavity is absent in acoelomates

Sol.: Given figure represents diagrammatic sectional view of coelomate animals.

The body cavity which is lined by mesoderm is called coelom. Animals possessing coelom are called coelomates, *e.g.*, annelids, molluscs, arthropods, echinoderms, hemichordates and chordates.

In pseudocoelomates, the body cavity is not lined by mesoderm. Instead, the mesoderm is present as scattered pouches in between the ectoderm and endoderm. *E.g.*, Aschelminthes.

178. Answer (2)

Hint: Larval form of echinoderms

Sol.: Coelenterates and ctenophores exhibit radial symmetry. Platyhelminths are the first group of animals that show bilateral symmetry. From platyhelminths onwards, organisms exhibit bilateral symmetry.

Adult echinoderms are radially symmetrical whereas larval form of echinoderms are bilaterally symmetrical.

Meandrina – Coelenterate

Asterias, Ophiura – Echinoderms

Ctenoplana – Ctenophore

179. Answer (2)

Hint: Feature of aschelminths

Sol.: Roundworms are bilaterally symmetrical, triploblastic and pseudocoelomate animals. Alimentary canal is complete with a well-developed muscular pharynx. An excretory tube removes body wastes from the body cavity through the excretory pore.

Malpighian tubules are excretory structures present in arthropods. Statocyst is present in arthropods.

Body is covered by a calcareous shell in molluscs.

180. Answer (4)

Hint: Coelenterates and ctenophores

Sol.: An undifferentiated layer, mesoglea, is present in between the external ectoderm and internal endoderm in diploblastic animals, seen in coelenterates and ctenophores.

Coelenterates – *Physalia*, *Adamsia*, *Gorgonia*, *Meandrina*

Ctenophore – *Pleurobrachia*

Molluscs – *Aplysia, Dentalium, Chaetopleura*

Echinoderm – *Antedon*

Both molluscs and echinoderms are triploblastic animals.

181. Answer (3)

Hint: A thin and flexible membrane.

Sol.: The entire body of cockroach (*Periplaneta americana*) is covered by a hard chitinous exoskeleton (brown in colour). In each segment, exoskeleton has hardened plates called sclerites (tergites dorsally and sternites ventrally) that are joined to each other by a thin and flexible articular membrane (arthrodial membrane).

182. Answer (1)

Hint: Transparent and membranous wings

Sol.: The first pair of wings arises from mesothorax and the second pair of wings arises from metathorax. Forewings (mesothoracic) called tegmina are opaque dark and leathery and cover the hindwings when at rest. The hindwings are transparent, membranous and are used in flight.

183. Answer (4)

Hint: Aquatic chordate

Sol.: Heart is ventral and CNS is dorsal, hollow and single in chordates. *Balaenoptera* (Blue whale) is a mammal and belongs to the phylum Chordata.

Aplysia, *Fasciola* and *Anopheles* are non-chordates. Heart is dorsal and CNS is ventral, solid and double in non-chordates.

184. Answer (2)

Hint: An aquatic annelid

Sol.: *Nereis*, *Pheretima* and *Hirudinaria* belong to the phylum Annelida. *Nereis*, an aquatic form, is dioecious, but earthworm (*Pheretima*) and leech (*Hirudinaria*) are monoecious.

Balanoglossus belongs to the phylum Hemichordata and is unsegmented.

185. Answer (3)

Hint: Characteristic feature of spiny bodied animals.

Sol.: Water vascular system is present in echinoderms. Water canal system is present in poriferans. Water enters through minute pores (ostia) in the body wall into a central cavity, spongocoel, from where it goes out through the osculum.

Choanocytes or collar cells line the spongocoel and the canals.

SECTION-B

186. Answer (3)

Hint: Labium is unpaired

Sol.: In cockroach, the mouthparts consist of a labrum (upper lip), a pair of mandibles, a pair of maxillae and a labium (lower lip).

Respiration takes place by network of tracheae. Tracheae opens outside with spiracles.

There are ten segments in abdomen.

Determinate and radial cleavage takes place during embryonic development.

187. Answer (1)

Hint: Cockroach has nocturnal vision.

Sol.: In cockroach, each eye consists of about 2000 hexagonal ommatidia. With the help of several ommatidia, a cockroach can receive several images of an object. This kind of vision is known as mosaic vision with more sensitivity but less resolution, being common during night (hence called nocturnal vision).

188. Answer (4)

Hint: Opens into the genital chamber

Sol.: Cockroaches are dioecious and both sexes have well-developed reproductive organs.

Male reproductive system consists of a pair of testes, one lying on each lateral side in the 4th to 6th abdominal segments. A characteristic mushroom-shaped gland is present in the 6th to 7th abdominal segments which functions as an accessory reproductive gland.

The female reproductive system consists of two large ovaries, lying laterally in the 2nd to 6th abdominal segments. A pair of spermatheca is present in the 6th segment which opens into the genital chamber.

189. Answer (2)

Hint: Endorphins are pain relievers produced by the brain

Sol.: Endocrine cells present in different parts of the gastro-intestinal tract secrete four major peptide hormones; namely gastrin, secretin, CCK and GIP.

Gastrin acts on the gastric glands and stimulates the secretion of HCl and pepsinogen.

Secretin acts on the exocrine pancreas and stimulates the secretion of water and bicarbonate ions.

CCK acts on both pancreas and gall bladder and stimulates the secretion of pancreatic enzymes and bile juice, respectively. GIP inhibits gastric secretion and motility.

Endorphins are neuropeptides that block the perception of pain.

190. Answer (3)

Hint: Catecholamines

Sol.: Antagonistic hormones produce opposite effects.

The adrenal medulla secretes two hormones called adrenaline or epinephrine and noradrenaline or norepinephrine. These are rapidly secreted in response to stress of any kind and during emergency situations.

Glucagon increases blood glucose levels whereas insulin decreases blood glucose levels.

Aldosterone leads to an increase in blood pressure whereas ANF can cause vasodilation and thereby, decrease the blood pressure.

PTH is a hypercalcemic hormone. Along with TCT, it plays a significant role in calcium balance in the body.

191. Answer (4)

Hint: Thymus

Sol.: The thymus plays a major role in the development of the immune system. This gland secretes the peptide hormones called thymosins. Thymosins play a major role in the differentiation of T-lymphocytes, which provide cell-mediated immunity. In addition, thymosins also promote production of antibodies, to provide humoral immunity. Thymus is degenerated in old individuals.

192. Answer (3)

Hint: ADH acts on DCT

Sol.: Vasopressin, released by neurohypophysis acts on kidneys and stimulates reabsorption of water and electrolytes by the distal renal tubules and thereby reduces the loss of water through urine (diuresis). Hence, it is also called anti-diuretic hormone (ADH).

193. Answer (3)

Hint: Neck and tail are absent in frogs.

Sol.: Body of frog is divided into head and trunk. Tail is absent. The hindlimbs end in five webbed digits and they are larger and muscular than forelimbs that end in four digits.

Skin of frog is always in a moist condition. Scales are absent.

The colour of dorsal side of the body is generally olive green with dark irregular spots. On the ventral side, the skin is uniformly pale yellow.

Heart of frog is myogenic.

194. Answer (1)

Hint: Ventricle of frog pumps out mixed blood.

Sol.: In frog, the heart is three-chambered, two atria and one ventricle and is covered by a membrane called pericardium. A triangular structure called sinus venosus joins the right atrium. It receives blood through major veins called vena cava. The ventricle opens into a sac-like conus arteriosus on the ventral side of the heart.

195. Answer (2)

Hint: Cyclostomes are jawless vetebrates.

Sol.: Agnatha and Gnathostomata are divisions of subphylum Vertebrata. Cyclostomata is a class under division Agnatha. Pisces and Tetrapoda are superclasses of division Gnathostomata.

196. Answer (3)

Hint: Additional chambers in digestive tract

Sol.: The characteristic feature of birds are the presence of feathers. The forelimbs are modified into wings. The hindlimbs generally has scales and are modified for walking, swimming or clasping the tree branches.

Skin is dry without glands except the oil gland at the base of the tail. Endoskeleton is fully ossified (bony) and the long bones are hollow with air cavities (pneumatic).

The digestive tract of birds and cockroaches have additional chambers, the crop and gizzard.

197. Answer (1)

Hint: Secreted by corpus luteum

Sol.: Progesterone, a steroid hormone supports pregnancy. Progesterone also acts on the mammary glands and stimulates the formation of alveoli (sac-like structures which store milk) and milk secretion.

Estrogen stimulates growth and development of female accessory sex organs. Prolactin regulates the growth of the mammary glands and formation of milk in them. Oxytocin acts on the smooth muscles of human female body and is a milk ejecting hormone.

198. Answer (2)

Hint: Acts on the melanocytes

Sol.: Pars intermedia secretes only one hormone called melanocyte stimulating hormone (MSH).

ACTH stimulates the synthesis and secretion of steroid hormones called glucocorticoids from the adrenal cortex. Both LH and FSH stimulate gonadal activity and are called gonadotropins.

In males, LH stimulates the synthesis and secretion of hormones called androgens from testis. In females, LH induces ovulation of fully mature follicle (Graafian follicles) and maintains the corpus luteum.

199. Answer (2)

Hint: Has flame cells for excretion

Sol.: In poriferans, (*e.g.*, *Sycon*), cells are arranged as loose cell aggregates.

Tissue level of body organisation is seen in coelenterates (*e.g.*, *Pennatula*).

Incomplete digestive system means presence of a single opening which acts as both mouth and anus. Platyhelminths (*e.g.*, *Planaria*) has incomplete digestive system.

Enterobius vermicularis is an oviparous aschelminth.

200. Answer (4)

Hint: Function of MSH

Sol.: The pineal gland is located on the dorsal side of the forebrain.

Pineal gland secretes a hormone called melatonin.

Melatonin plays a very important role in the regulation of 24-hour rhythm of our body. It also helps in maintenance of body temperature.

In addition, melatonin also influences metabolism, pigmentation, the menstrual cycle as well as our defense capability. A hormone called MSH secreted from pars intermedia also regulates pigmentation.

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